Where does Qwen3-32B hold a paper's authors?

Stages 1–9

Small samples: find a signal, try to break it

20 → 50 → 200 papers (Stage 4, Pilot50, Stage 8 cohorts)

Stages 10–13 · post-defence additions (2026-11-05) at the end

Full cohort: locate it, then intervene

9,108 papers · 149,999 name tokens (Stage 10 cohort: papers.parquet, name_tokens.parquet)

One question, four lines of argument

Each line narrows the space of explanations; small samples set it up, the full cohort settles it

dark pages = Stages 1–9 · light pages = Stages 10–11 · purple tag = comparison

Where does Qwen3-32B hold a paper's authors, and can it say them?

A · This paper?

Paper-specific, not an easy name?

- 1–9 · Stage 5, 7.2, 7.3, 8
- 10–11 · title signal, name prior, probe

B · Readout tool

What can each readout see?

- 1–9 · Stage 3, 7.4, 8, 9
- 10–11 · lens comparison, linear probe

C · Which layer

When does it appear, when is it used?

- 1–9 · Stage 4, 7.2, 7.3, 8 (L56)
- 10–11 · fine layers, patching, knockout

D · Which token

Which part of name or prompt carries it?

- 1–9 · Stage 6, 7.1
- 10–11 · name format, tail tokens, co-authors

Stages 1–9 · 20–200 papers

Pilot design

- Pilot50 / Stage 8: 50 and 200 papers (pilot50_v1.csv, stage8_papers.csv)
- Mostly first-listed author, first token
- Score: gold vs 1 or 100 false names (logit margin, win rate)
- A/B both R0, matched in pairs
- Wrong titles: 1 crossed or 20 shuffled
- Layer L56 chosen after repeated looks

Stages 10–11 · 9,108 papers

Full-cohort design

- All 9,108 papers (papers.parquet), every author, every token
- Score: rank over the whole vocabulary, log10(1/rank)
- target = A + C, control = B + D (R0 + R1), compared within citation bins
- 5 matched false titles + no title
- All layers reported, fine layers added
- Parity and Stage 6 cross-check before analysis

What changed in the design, and why

Small samples could not separate "small effect" from "no effect" in a given layer. The full cohort trades candidate-based scores for full-vocabulary ranks and pays for it with GPU time. **Note:** target/control include R1 papers; with R0 only (A vs B, as in the pilot) the output gap is +0.199 vs +0.207 (Stage 10, final L63, R0 3,423 vs 2,151; §6.13) and every layer looks the same.

Same papers, two ways of grouping

Stages 1–9 compare A vs B; Stages 10–11 compare target vs control. Counts from the 6,841-paper cohort.(Stage 1, 351_audit.json group_counts)

	recognised (target)	not recognised (control)
R0 · no true author written	A = 4,110	B = 2,186
R1 · some authors written (HR2 > 0.5)	C = 511	D = 34
Total	**target = A + C = 4,621**	**control = B + D = 2,220**

Stages 1–9

A vs B: both R0, so the contrast is purely "knows but cannot say"

Stages 10–11

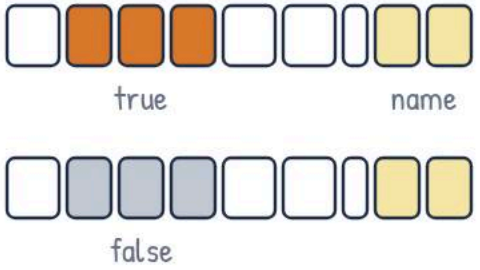
(A + C) vs (B + D): adds 511(351_audit.json) partial-recall papers, almost all on the target side

Check done: with R0 only (A vs B) the output gap is +0.199, versus +0.207 for target vs control(Stage 10, final L63, within-bin; §6.13), and every layer looks the same. The two phases compare like with like.

Reading the method figures

1 2 3 steps, in order

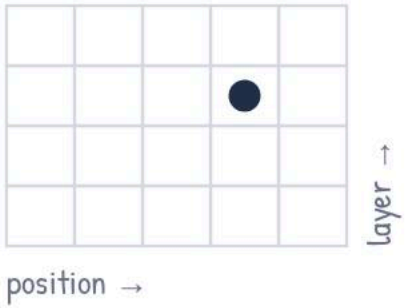
Prompt tokens



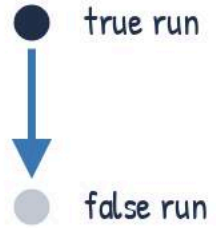
Layers



State = dot



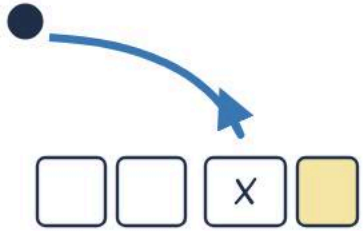
Copy a state



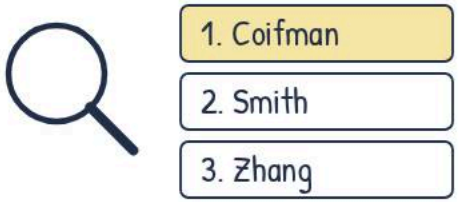
Block



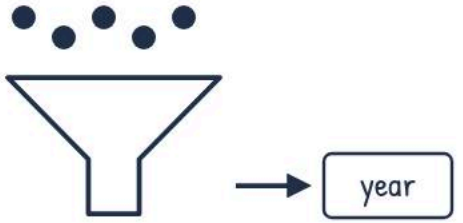
Inject



Read out



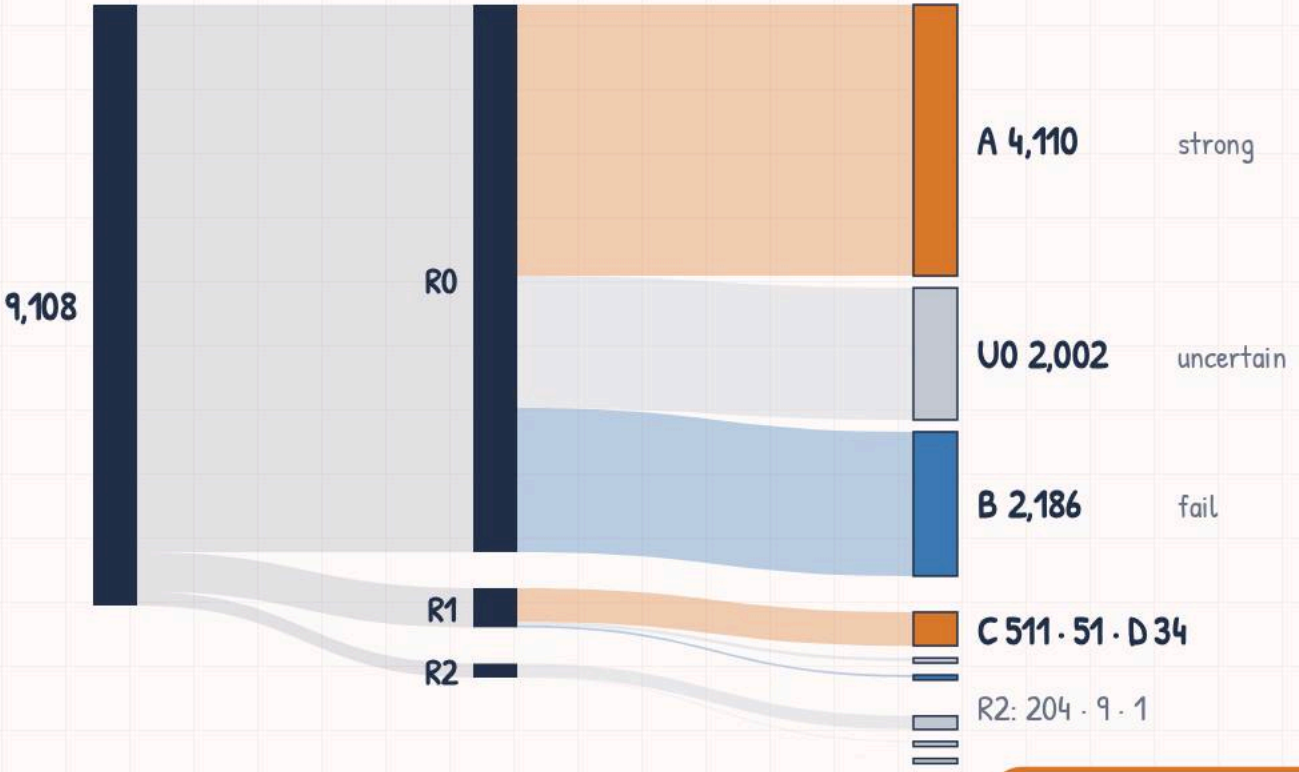
Probe



Groups and samples

example: cell C

1 papers → 2 open recall → 3 recognition → 4 cells → 5 samples



Stage 10 · M2 · M3

all papers

Stages 11–13 · X

593 target + 593 control

1,178 matched pairs

Stages 1–9

Pilot50 · Stage 8 (A, B)

target = A + C = 4,621 · control = B + D = 2,220

Counts: 351_stage10/cohort/papers.parquet (recall_level × MC outcome); 593 + 593: 361_stage11 sample (1,186 papers); 1,178 pairs: Stage 13 X1/X3 pair table

Cohort and behavioural labels

From the 15 Oct label package (9,108 papers) to a frozen 6,841-paper analysis cohort (351_audit.json total, four_group_total)

Group	Recognition / open recall	Full (351_audit.json)	Pilot50 (pilot50_v1.csv)
A	pass / R0 (no true author written)	4,110	20
B	fail / R0	2,186	20
C	pass / R1 (some authors matched)	511	7
D	fail / R1	34	3

- Pass = 4/4 option orders correct *and* mean true-option probability ≥ 0.5; fail = 0–1/4
- Pilot50: seed 35150, keeps the 20 Stage 4 papers + 30 new
- 20 A/B pairs matched on exact field and citation bin, then $|\Delta\text{year}| + 2 \cdot |\Delta\text{authors}|$

Why A vs B?

Both groups write **no** correct author. They differ only in whether the model recognises the true list. That isolates "knows but cannot say".

Caveat

In the 0–2-citation bin the strict pass rate is **25.5%** (796 papers; label_README.md l.8), above the planned ≤10% floor: "pass" is not proof of stored knowledge.

One fixed model, prompt and score

Everything later is measured under these settings

Model	Qwen/Qwen3-32B, BF16, revision 9216db57, 64 blocks, d = 5,120
Layers	L16, 24, 32, 40, 48, 56, 62 → target L63
Readouts	Public J-Lens · Vanilla (logit lens) · real final logits
Position	last input position; context limit 512
Names	encode(prompt) + encode(" " + author)
Hardware	29 blocks on GPU, 35 offloaded to CPU

> Who are the authors of the paper titled "{title}" ({year})?
> Authors:

Rank r = 1 + #tokens with a strictly larger logit

JMRR = mean over authors of mean over tokens of 1/r

Integrated = average of the 7 source layers (L63 excluded)

Guard: forced token IDs are checked, and a J-Lens/Vanilla final-logit gap > 0.1 rejects the run.

How groups and contrasts are built

labels: group lead's pipeline on the same Qwen3-32B · we use them unchanged

open answer -> R0 / R1 / R2

greedy, chat template

fuzzy name match: M, H, U

HR2 = (H+U) / (M+H+U)

R2: HR2 <= 0.5 · R0: M = 0 · R1: M > 0

a fixed rule, no human or model judge

multiple choice -> target / control

true list + 3 real teams of other papers

(same field, team size, citation bin, ~year)

4 orders · p(letter) from next token

target: 4/4 and mean p(true) >= 0.5

control: <= 1/4

5 false titles per paper

same field, own year kept

min-cost on |dyear|/5 + |dauthors|

+ |dtitle tokens| + 1.5 |dcitation bin|

no shared surname, Jaccard <= 0.6

79% strict tier (Stage 10, 35,898 of 45,540 pairs, conditions_manifest.json)

statistics

target vs control only within citation bin

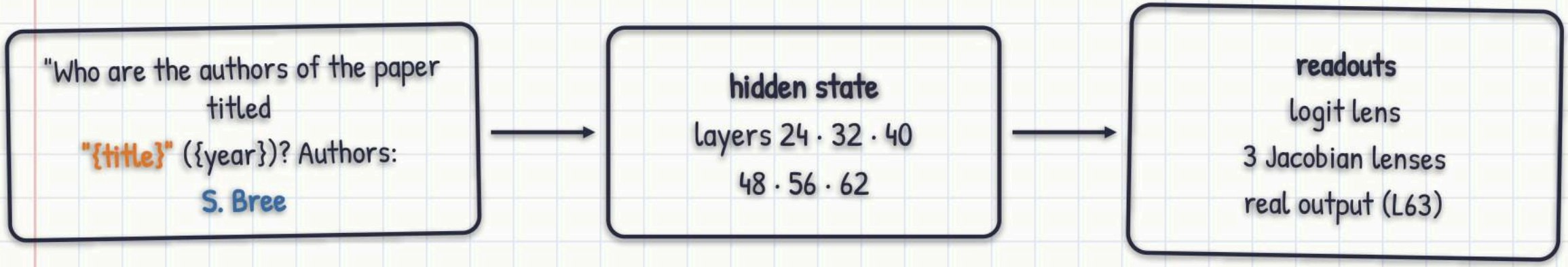
10 bins with >= 30 per group (Stage 10, §6.2)

weight $n_t n_c / (n_t + n_c)$

95% bootstrap over papers, 2,000

sensitivity: drop bins 0-4

How we read the model



Score = rank of the correct next name token,

as log10(1 / rank): 0 = rank 1, -1 = rank 10

7 title conditions: the true title, 5 matched false titles, no title

title-specific signal

= true title - mean(5 false titles)

Same name, same tokens: what is left is knowing

this paper

Lenses (model weights never change; Jacobians to L63 averaged over prompts): public = 557 WikiText-103 prompts (Neuronpedia) · self100 = 100 frozen WikiText-103 prompts · fitA = 100 other WikiText-103 prompts (Stage 9: one skipped prompt replaced -> 100 valid) (n_prompts: 351_stage10/config/lenses.json; Stage 9 final_status.json)

Can we trust the numbers?

fast = slow

One forward pass vs one per prefix: max gap 0.37 logit (bf16 rounding); ranks 1-41 identical (Stage 10, 12 contexts, parity.json)

matches Stage 6

972 tokens: same token ids, median log-prob gap ~1e-5 (Stage 10, crosscheck_stage6.csv)

negative control

Punctuation tokens show ~0 title-specific signal at every layer (Stage 10, max 0.014, 9,108 papers, §6.5)

Caveat: the parity tolerance was raised 0.25 -> 0.5 after seeing these bf16 gaps. (parity.json tolerance_amendment)

Line A

Is it about this paper?



small samples asked

Does the true title beat a crossed or shuffled title? Does A gain more than B?

→

the full cohort answered

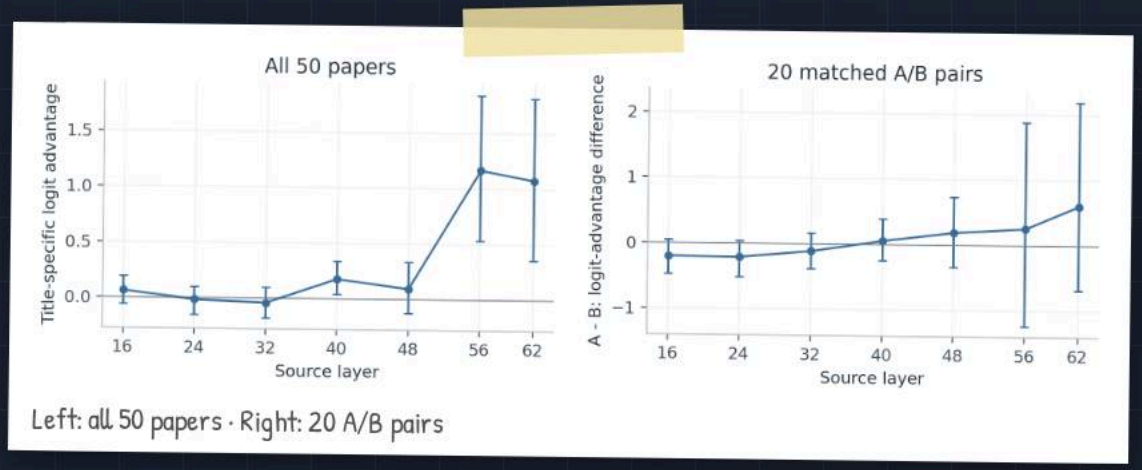
Title-specific signal for every token; target beats control only late; name prior split out; probe controls

Crossed title–author control

Is the gold name high because of the title, or because the name is easy?

Paper P (author G) and source paper Q (author F, same field & bin).
M = logit(G) − logit(F) · T = M(P title) − M(Q title)
A fixed preference for G cancels out.

- All 50 papers: L56 +1.17 [0.53, 1.84], L62 +1.08 [0.36, 1.82] (Stage 5, J-Lens, 50 papers, layer_summary.csv)
- Matched A–B, 7-layer integrated: +0.08 [−0.43, +0.65] (20 pairs, matched_ab_summary.csv)

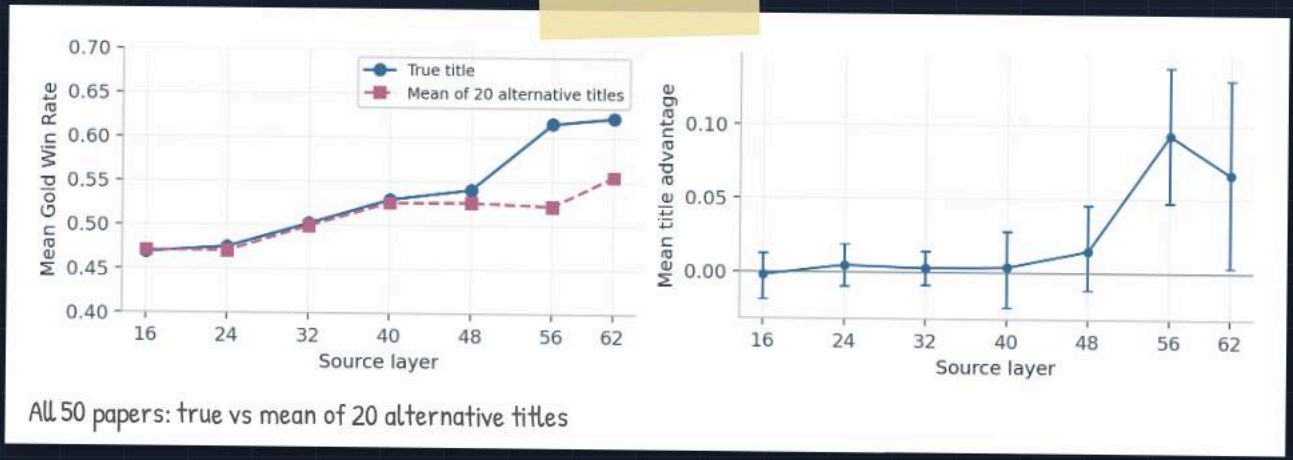


Left: all 50 papers · Right: 20 A/B pairs

The title changes which name wins in late layers. Whether A gains more than B: **not established**.

Shuffled titles: the true title helps

Same 100 candidates, 20 metadata-matched wrong titles per paper, 1,000 contexts; only the title changes



All 50 papers: true vs mean of 20 alternative titles

- Pooled L56: 0.616 vs 0.523 → +9.3 pp [4.8, 14.0], 35/50 papers positive (J-Lens L56, 50 papers; paper_null_distribution.csv; CI recomputed 10-30)
- Leave-one-out: +8.6 to +10.0 pp (stage7_l56_loo.csv)
- Vanilla L56 +10.6 pp, final output +7.2 pp (paper_null_distribution.csv)
- Matched A–B integrated: +0.018 [−0.021, +0.063] (20 pairs, integrated_ab_bootstrap.csv)

Two different questions: "does the title help?" **yes**; "does it help A more than B?" **not shown**.

Does the title effect survive the obvious objections?

Source-author overlap

199/1,000 wrong titles come from a paper whose author is in the false pool. Excluding them: +0.026 → +0.028.(quality.json; paper_null_distribution.csv)

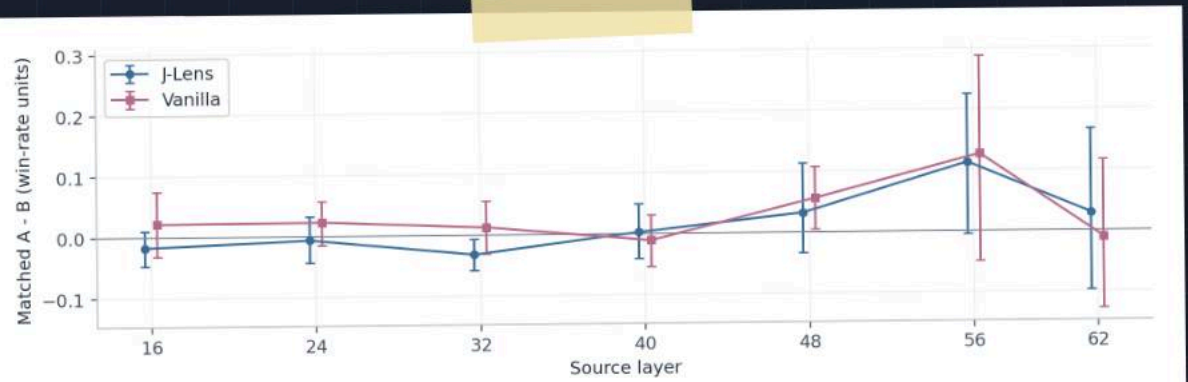
Candidate weighting

Author-, unique-token- and non-collision weights: pooled +0.026 to +0.032; all six A–B CIs include 0.

(stage7_weighting_title_advantage.csv)

Low citations

0–2 citations (n = 6): L56 +0.22, 5/6 positive, but 0/6 beyond its own 20-title tail (≤ 0.05).(floor_L56_low_six.csv)



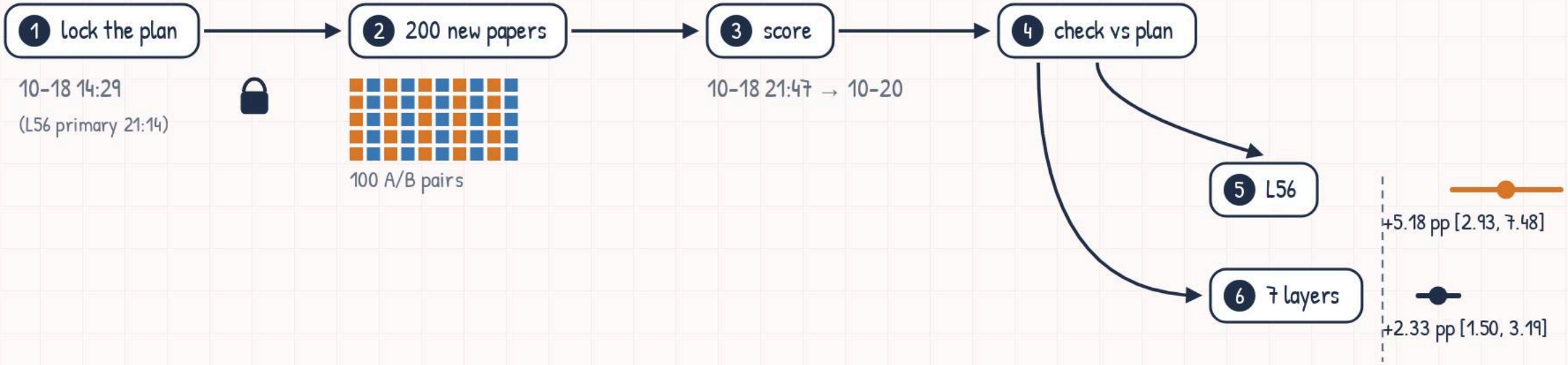
Title-adjusted A–B win rate by layer

Layer heterogeneity

L56 +0.11 but L32 **−0.03** (CI below 0).(20 pairs, matched_ab_bootstrap.csv) Reporting only the positive layers would mislead.

Independent replication (Stage 8)

plan first, data second

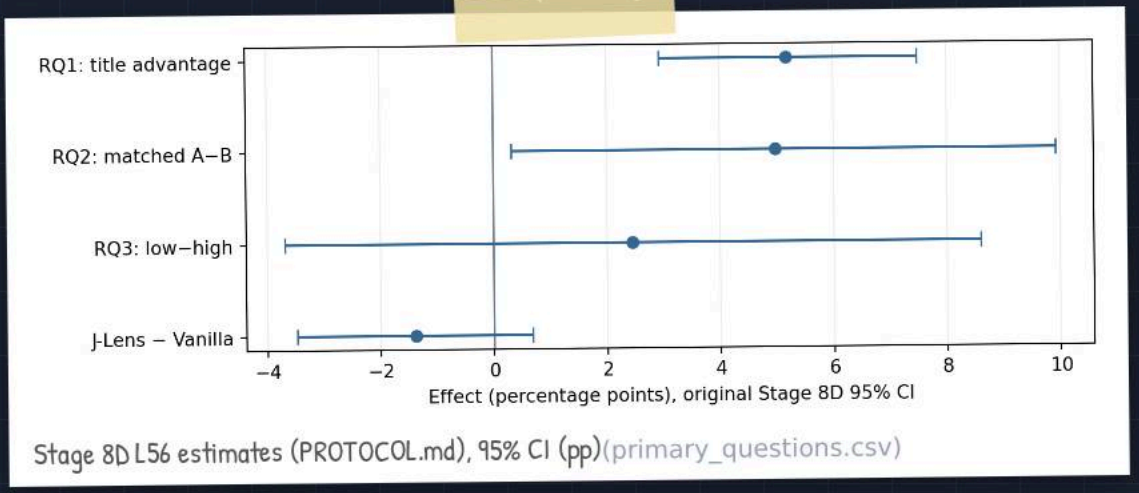


RQ1 = win rate(true title) − mean win rate(20 false titles)

first author vs 100 false names · J-Lens · paper bootstrap

+5.18 pp: Stage 8D RQ1, J-Lens L56, 200 papers, stage8D/summary/primary_questions.csv · +2.33 pp: J-Lens 7-layer, 200 papers, recompute_2026-10-30/s8_endpoints.csv

On 200 new papers the title effect replicates



Stage 8D L56 estimates (PROTOCOL.md), 95% CI (pp)(primary_questions.csv)

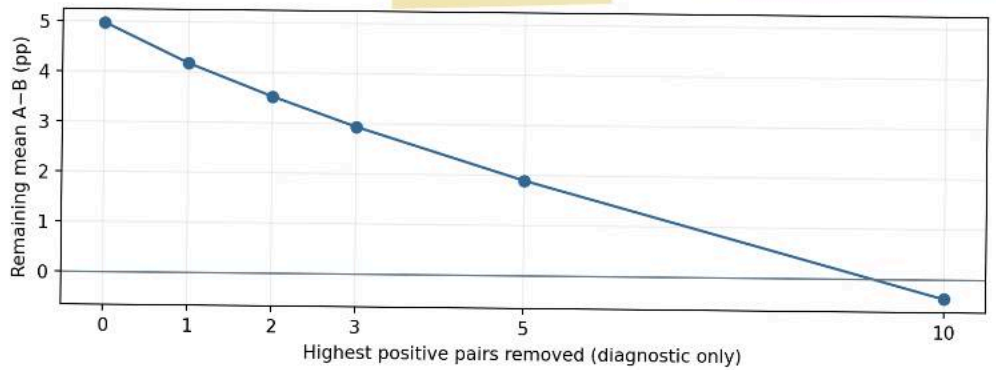
(Stage 8D, J-Lens L56; primary_questions.csv)	Estimate	95% CI
RQ1 title, L56(200 papers)	+5.18 pp	[+2.93, +7.48]
RQ2 A–B, L56(100 pairs)	+4.98 pp	[+0.32, +9.91]
RQ2 A–B, 7-layer (frozen primary)(100 pairs; s8_endpoints.csv)	+2.41 pp	[+0.91, +4.00]
RQ3 low–high(25 + 25 pairs)	+2.45 pp	[−3.69, +8.60]
J-Lens–Vanilla(jlens_minus_vanilla.csv)	−1.38 pp	[−3.48, +0.68]

Citation bands: low +6.8 [2.6, 11.6], mid +4.8, high +4.4 [−1.1, 10.0]

(50/100/50 papers; layer_and_band_summary.csv)

RQ1 is the first **independent** confirmation: correct titles help the gold author on new papers. J-Lens still adds nothing over Vanilla.

But the A–B gap rests on a few pairs



Remaining A–B after removing the largest positive pairs

- 100 pairs: 55 positive, 45 negative; median +4.4 pp (J-Lens L56; matched_pair_differences.csv)
- One pair (S8P094) ≈ +84 pp
- Remove top 5 → +1.9 pp; top 10 → −0.4 pp (recomputed from same file)

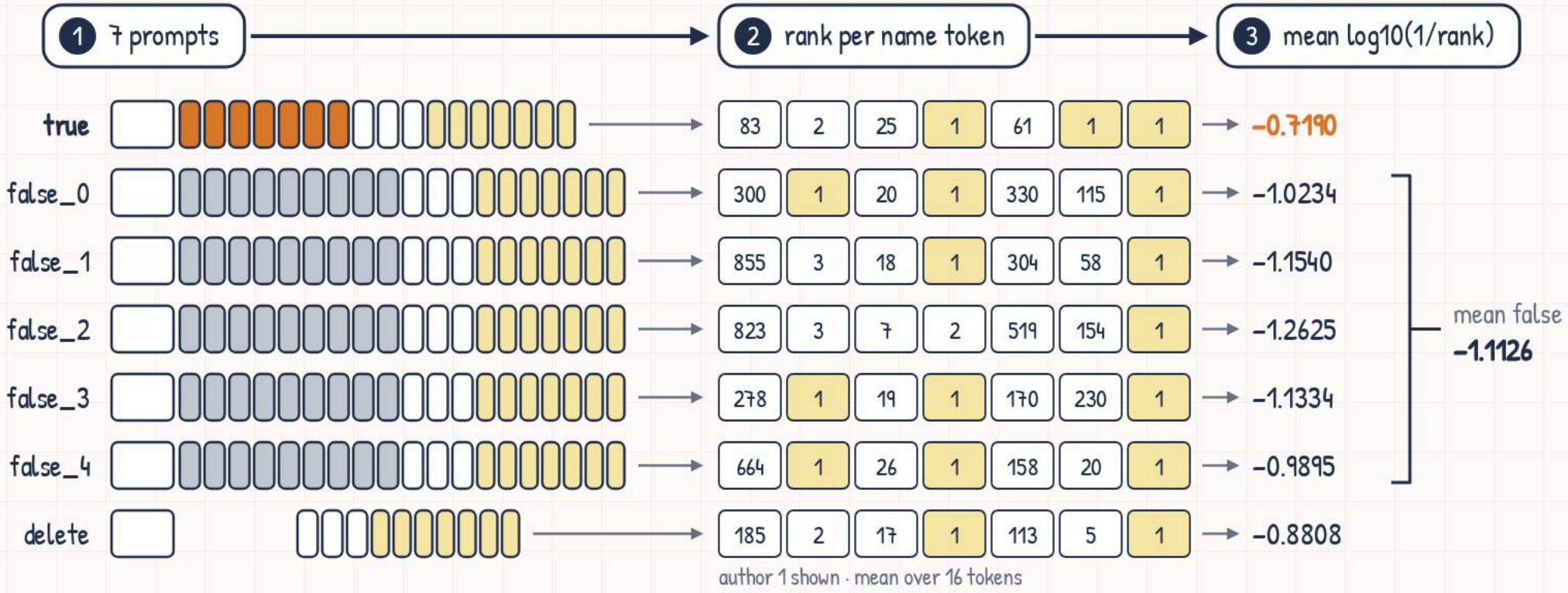
8E external audit (post-hoc)

- Unique-token weights: RQ1 +4.6, RQ2 +5.4 pp (s8_8E_weighting_by_paper.csv)
- Wrong-title length gap: +0.12 pp per token (200 papers; s8_unsourced_intervals.csv) (association only)
- Original scripts missing; recomputed from raw logits (30 Oct), same estimates (351_stage8/recompute_2026-10-30/): post-hoc only

Small samples can show a positive A–B mean that a handful of papers carry. This is the main reason Stage 10 moved to the full cohort.

Scoring: rank → log → spec

Coifman, Bonanno, Rafaeli (2007) · output layer



4 contrast

spec = mean nlr(true) − mean over 5 false titles

= −0.7190 − (−1.1126) = +0.3935

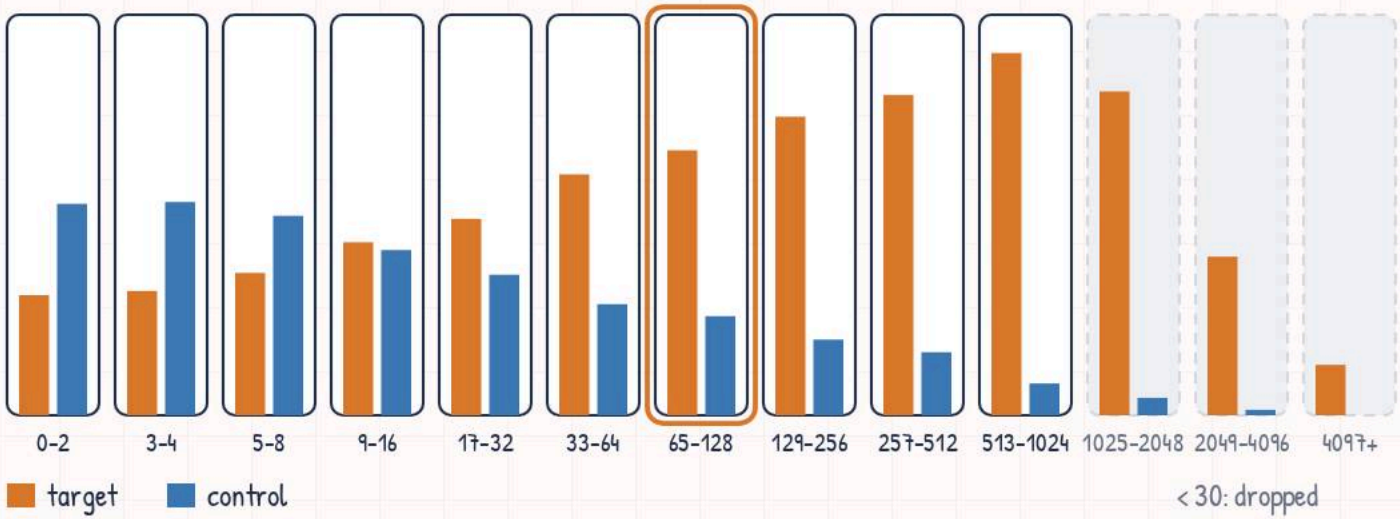
total = true − delete = +0.1618 · topic = mean false − delete = −0.2318

Ranks: 351_stage10/raw/[cond_*/]final/L63/shard_011.npz rows 25025–40 (author 1 shown) · nlr, spec, total, topic: summary_title/paper_title_contrasts.parquet (final, L63)

Statistical conventions

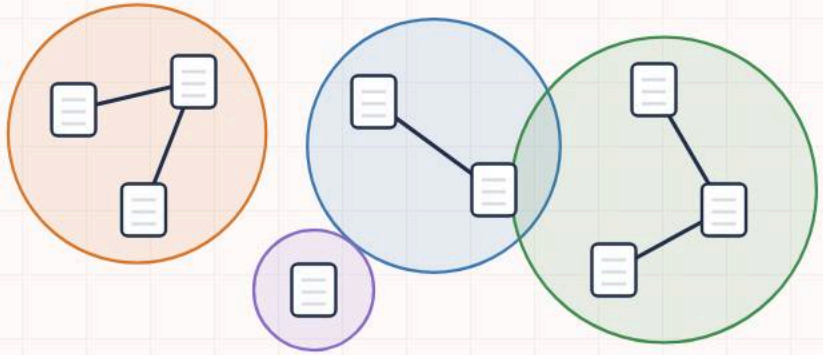
example paper: bin 65–128

1 citation bins → 2 target − control → 3 weighted mean



$$\Delta = \Sigma\, w_b \cdot (\text{mean}_t - \text{mean}_c),\ w_b \propto n_t \cdot n_c / (n_t + n_c)$$

4 teams → 5 resample teams



shared author name = one team

7,247 teams

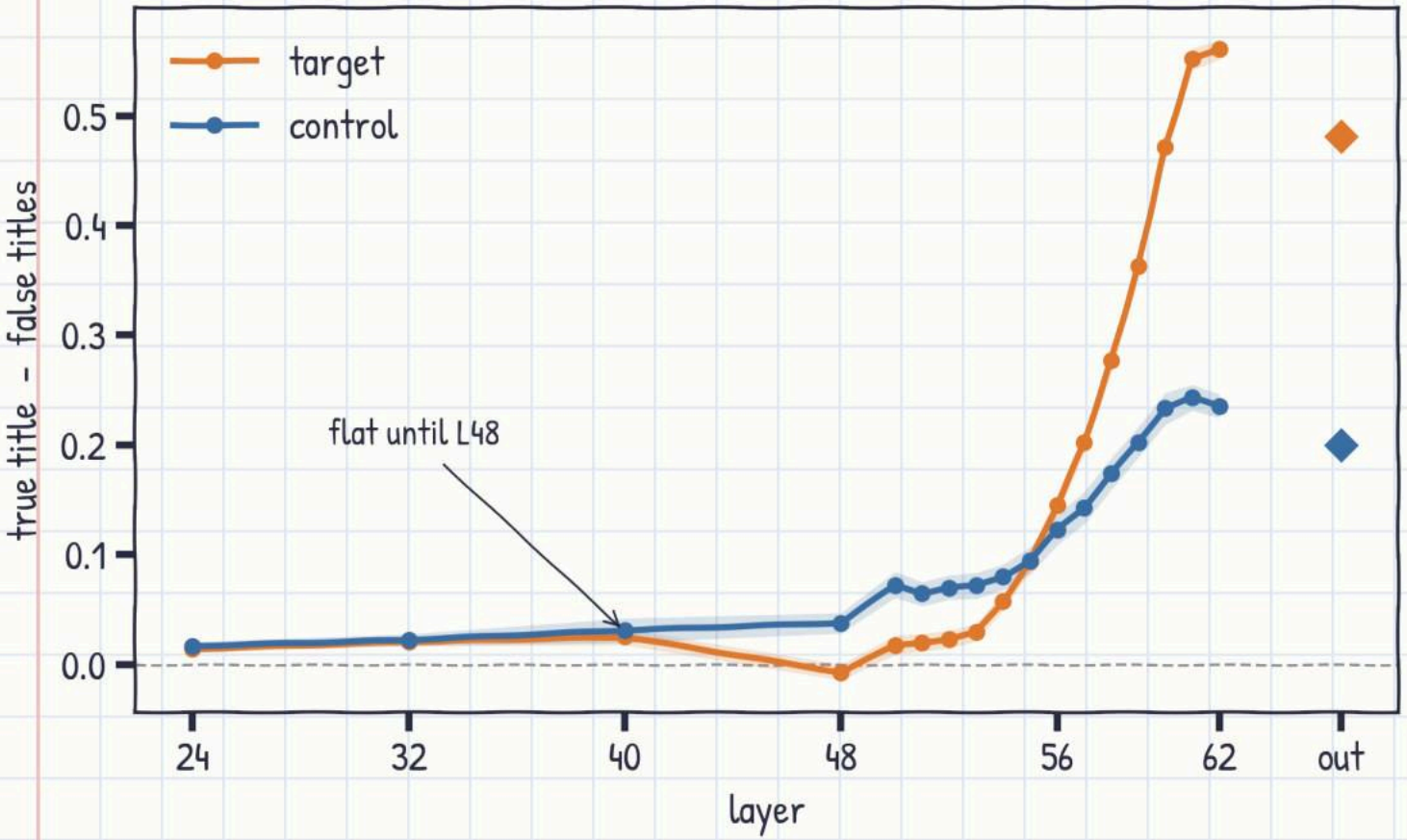


one team, one fold · 2,000 draws

Stage 10

Knowing the paper shows up only at the end

title-specific signal · public Jacobian lens (lines) and model output (diamonds)



L24–L48: ~0 for both groups (Stage 10, all readouts max 0.04, §6.3)

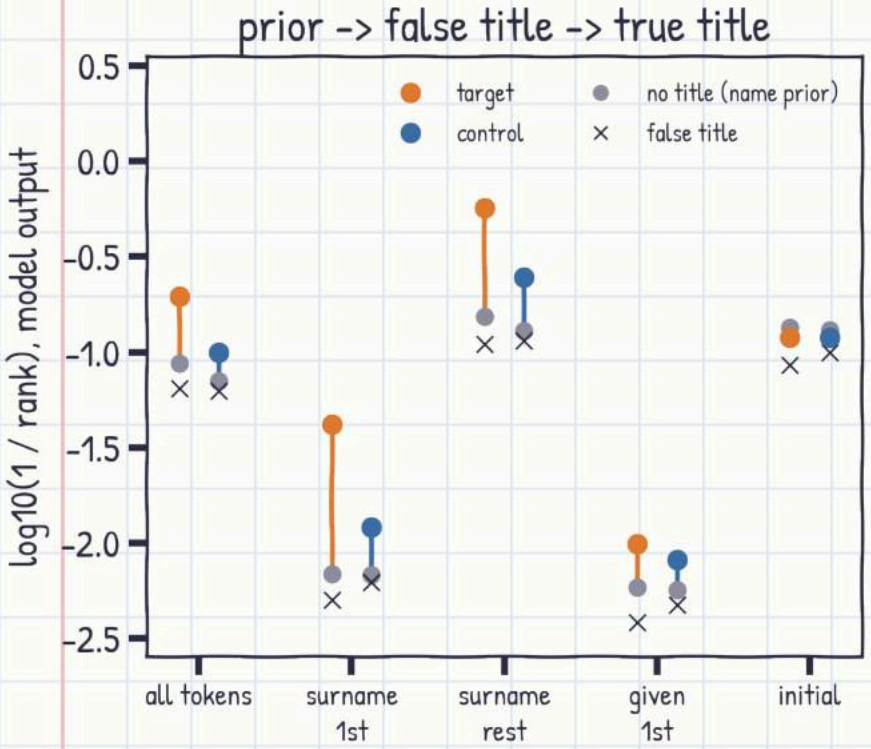
rises at **L56**

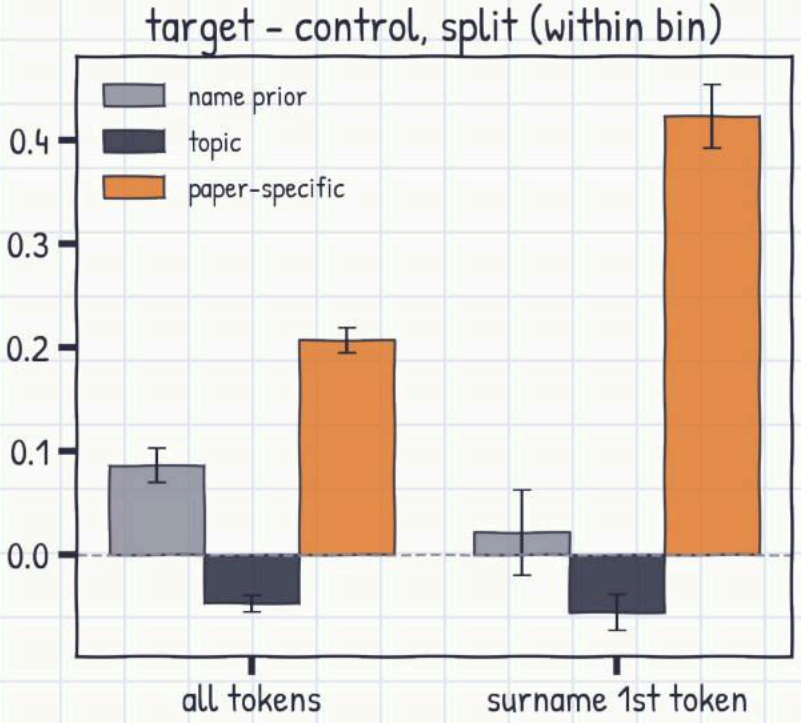
L62: **target +0.56** vs **control +0.24**
(Stage 10, public lens L62, 4,621 + 2,220 papers, §6.3 fig.)

Stage 11

How much is just the name prior?

model output · no title vs false title vs true title · all 9,108 papers





no title: rank ~12 (Stage 12 A, 9,108 papers, §6.12)

target - control: +0.09 prior, +0.21 paper (Stage 12 A, 4,621 + 2,220, within bin, §6.12)

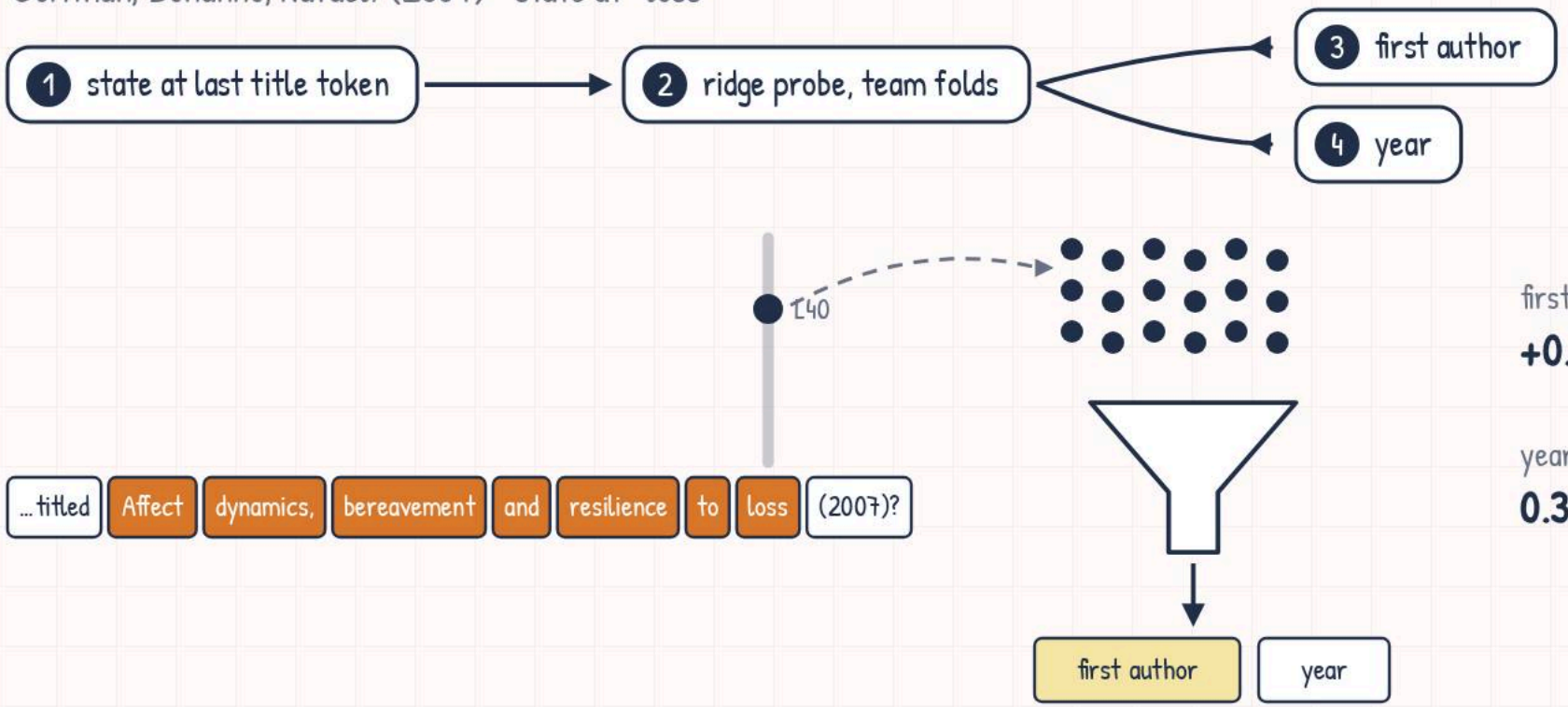
surname 1st: prior ~0, paper +0.42 (same, 1st surname token)

prior vs recognition rho 0.09 (Stage 12 B, 8,894 R0/R1 papers)

Probe

Coifman, Bonanno, Rafaeli (2007) · state at "loss"

⚠ scores only the first name token (mostly an initial) · all-token, by-role version in progress (addendum 9)



first author, L40

+0.061

year R^2, L16 → L52

0.38 → 0.56



colour = author team; one team, one fold

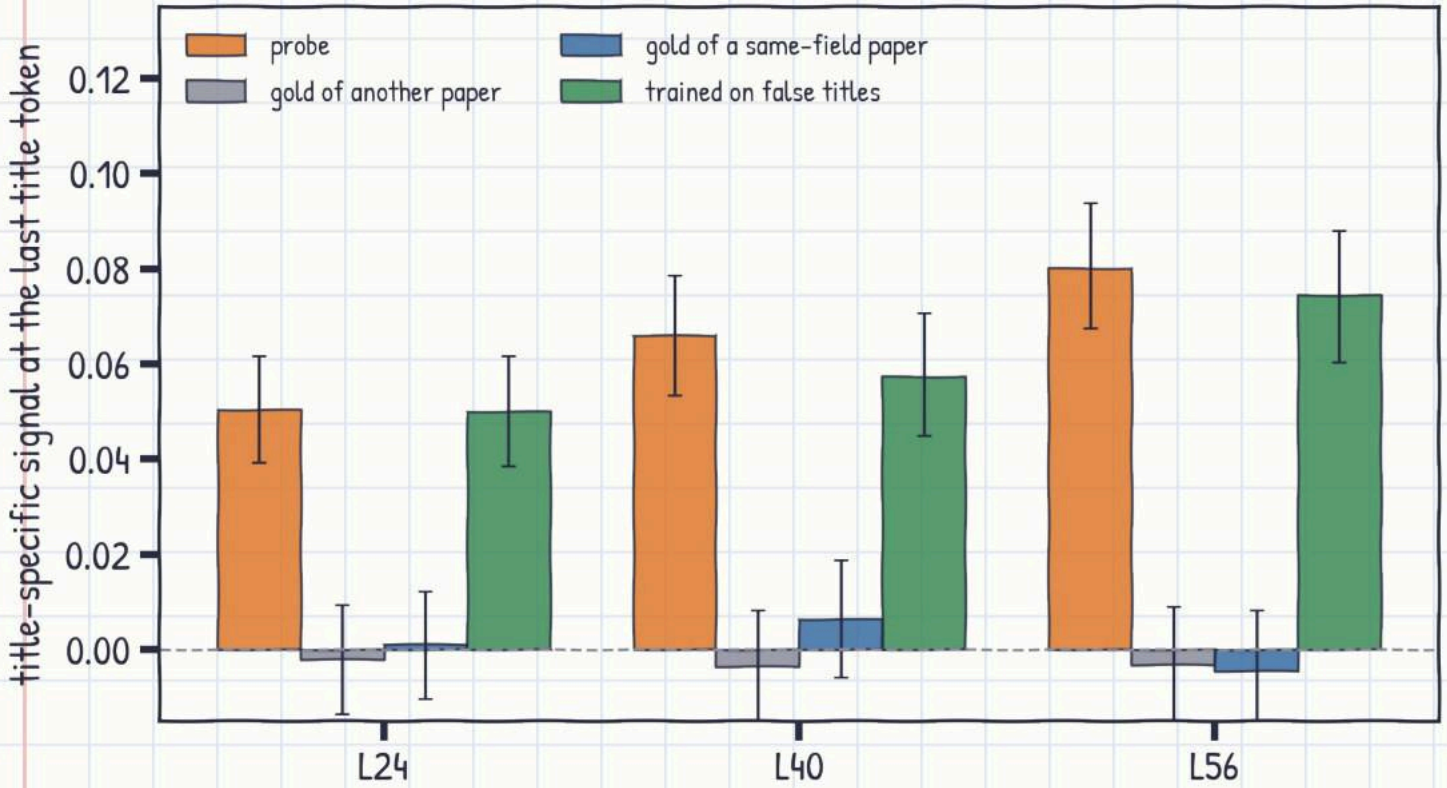
spec = probe(true title) − probe(false title)

probe score = log10(1/rank) of the true first-author token

+0.061: Stage 13 E4b re-check, probe scores ONLY THE FIRST NAME TOKEN, all papers (1,863), team folds, title reuse dropped; e4b_trainstd_compare.md
year R^2 0.38 → 0.56: Stage 13 M2, 9,108 papers, team CI; summary_cpu_teamci/M2_teamci.csv

The probe signal is paper-specific

last title token · 5-fold ridge probe · re-checked: author-team folds, title reuse removed



another paper's authors: **0**

same-field paper: **0** (Stage 12 X1 + Stage 13 re-check, §6.12)

so it is **this paper**, already at L24

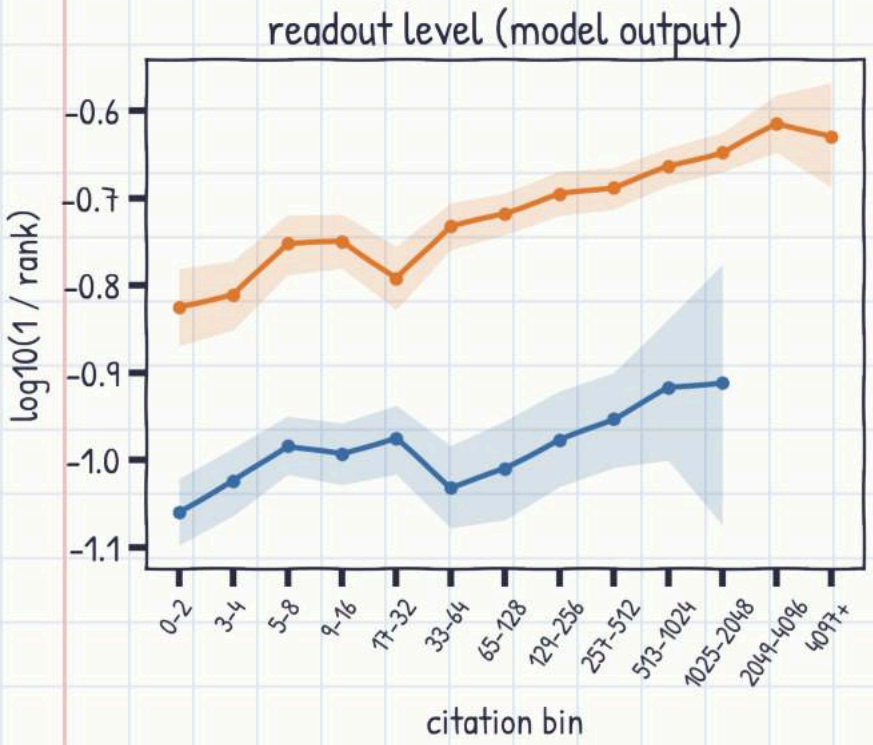
lenses cannot read it

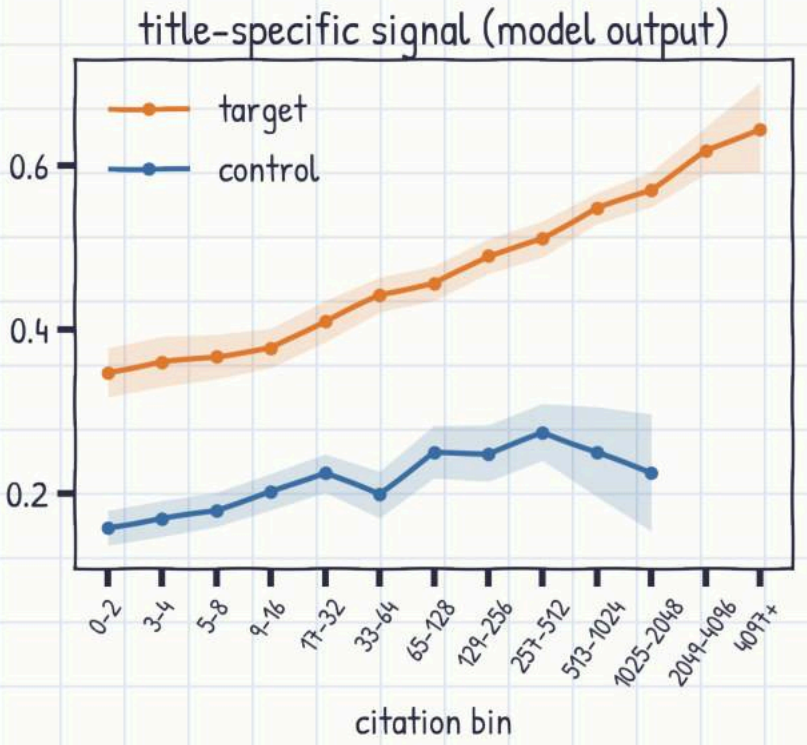
Probe scores only the first name token (mostly an initial); all-token, by-role version (addendum 9) in progress.

Stage 10

More citations, more paper knowledge

model output · bins with < 10 papers omitted





Target: +0.35 -> +0.64 (Stage 10, final, 200 / 84 papers, §6.8)

Spearman (title-specific)

0.45 all · 0.33 target · 0.19 control (Stage 10 X2, 9,108 papers, §6.8)

Target > control in every bin

Group comparisons

example paper: cell C

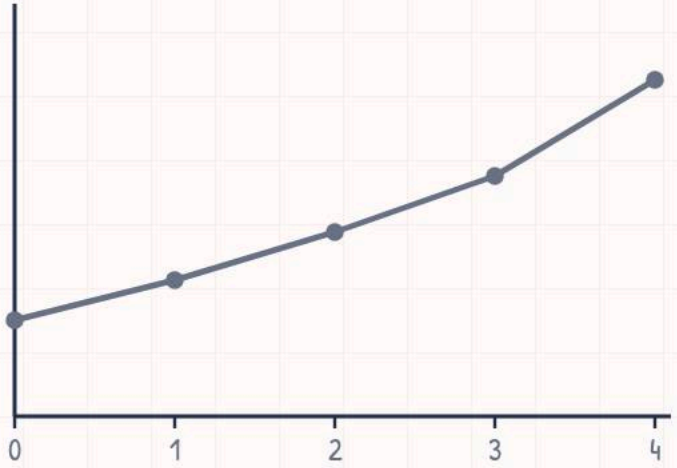
1. 9 cells
2. P1: A – B
3. P2: C – A

4. dose: orders right

P2 \ P1	strong	uncertain	fail
R0	A 4,110	U0 2,002	B 2,186
R1	C 511	51	D 34
R2	204	9	1

P1 Δ_1

P2 Δ_2



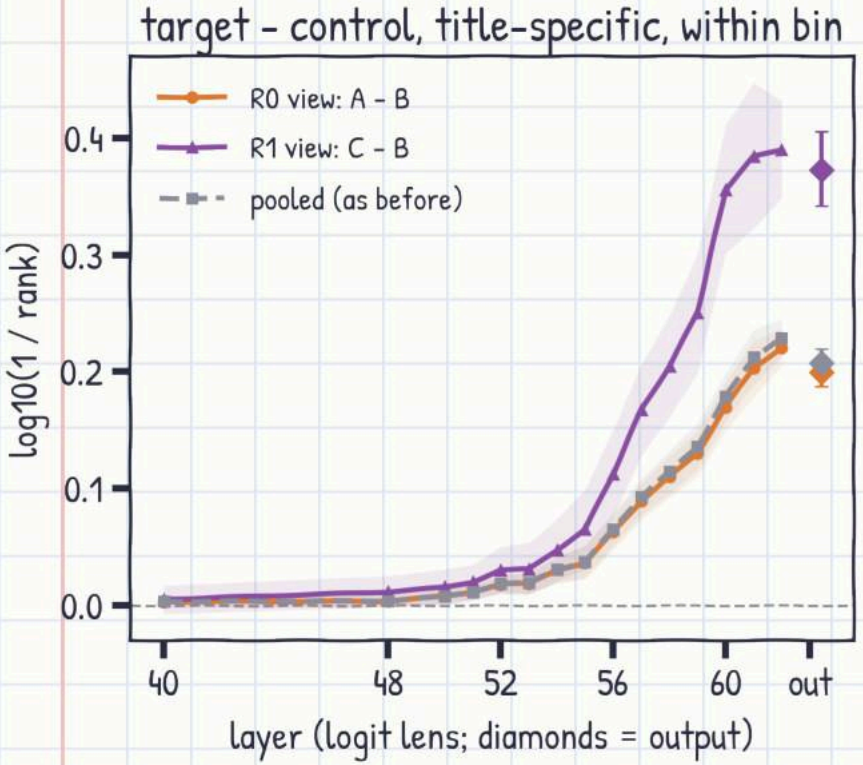
schematic, not results · 示意，不是结果

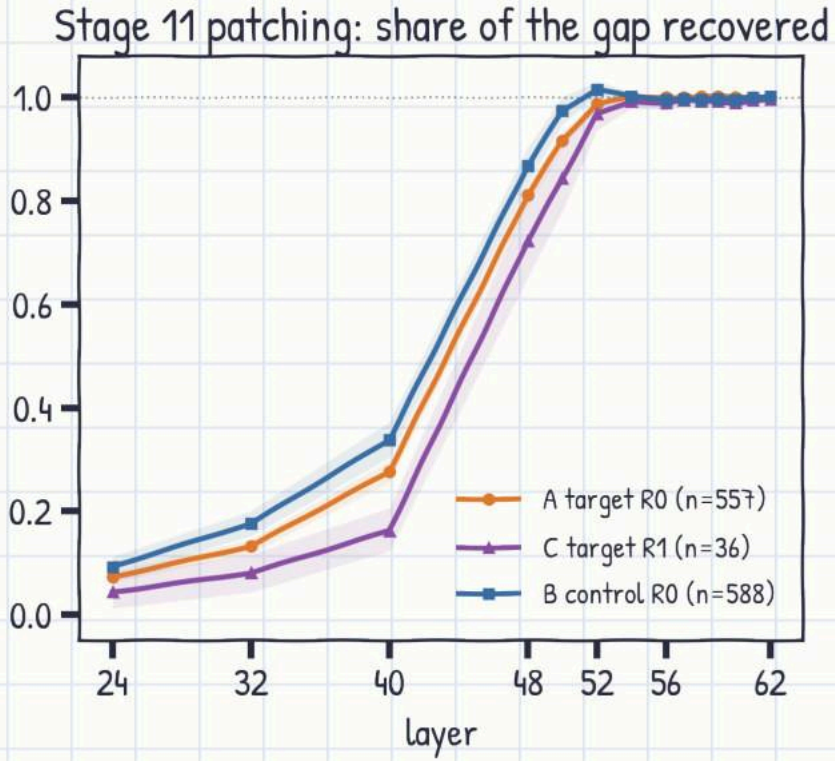
contrast = Σ w_b (mean_cell1 – mean_cell2)

title-specific signal, unwritten authors, output layer; team bootstrap

Is it just the R1 papers? No.

R0 = no true author written · R1 = a few written · A/B/C/D = target/control x R0/R1





R0 only (A – B):

output +0.199 vs +0.207, same shape (Stage 10, A 3,423 / B 2,151, §6.13)

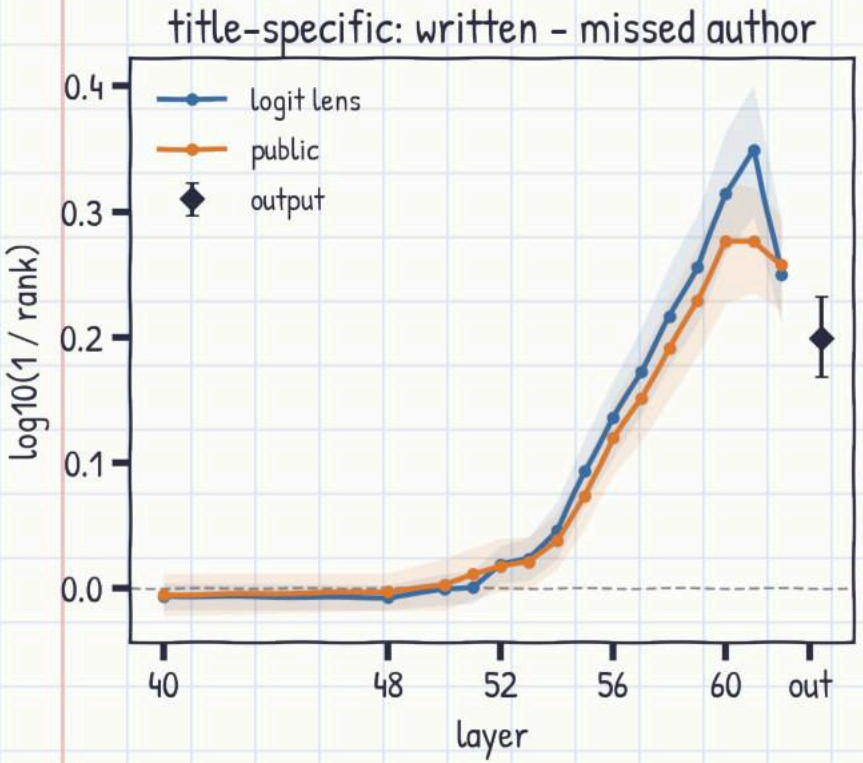
R1 (C – B): +0.37, also late (C 391 / B 885)

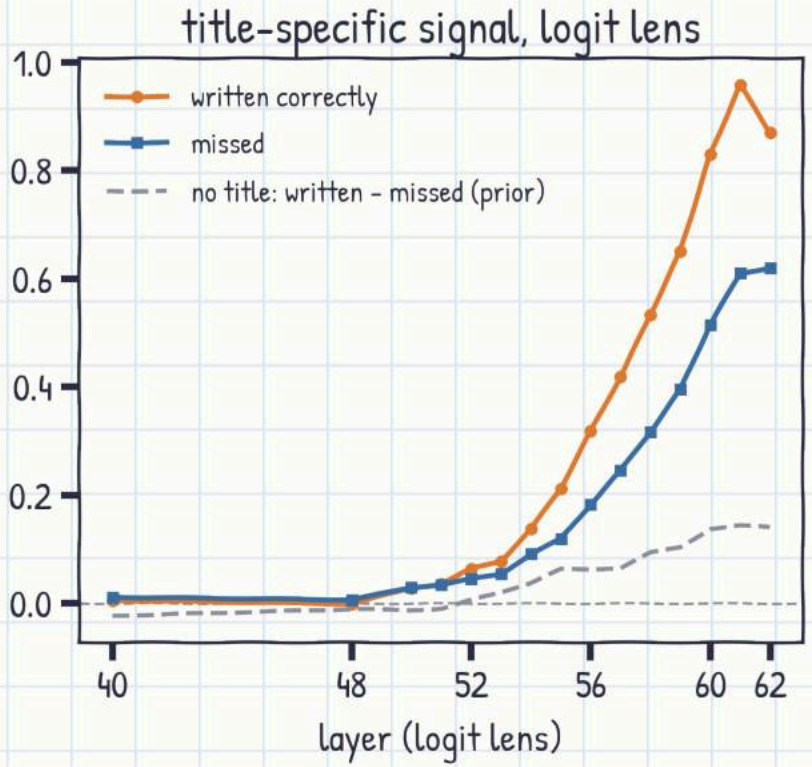
C – A within target: +0.11, from L56 (C 485 / A 3,204)

patching: R1 a bit later, done by L52 (Stage 11 E1, 36 R1 targets)

Authors it wrote vs authors it missed

inside 495 R1 papers · paired within paper · 95% CI (Stage 10, r0_r1_analysis.md §4)





written − missed, output:
+0.20 (Stage 10, final L63, 495 papers)

~0 up to L52, grows from **L54**

no title (prior): +0.10 (same, no title)

same without first authors

Stages 1–9 · 20–200 papers

What we saw

- True title helps: pilot L56 +9.3 pp (7.3, 50 papers); 200 new papers +5.2 pp (8D RQ1, 200 papers)
- A–B uncertain in the pilot; +5.0 pp on 200 papers but gone after dropping 10 pairs (8D RQ2, 100 pairs)
- Low – high citation: inconclusive

Stages 10–11 · 9,108 papers

What scale showed

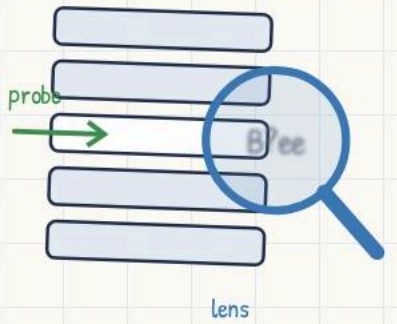
- Title-specific signal clear in every group, ≈0.44 at L62 (L62, §6.9)
- target – control ≈0 at L24–L40, slightly negative at L48 (−0.04 to −0.05) (§6.4), +0.21 [0.20, 0.22] at the output (L63, §6.4)
- Name prior is only ~1/3 of the gap; ~0 on surnames (§6.12)
- Signal rises steadily with citations (ρ = 0.45) (§6.8)
- Same groups as the pilot (R0 only, A – B): +0.20 [0.19, 0.21] (R0, §6.13), same late shape
- R1 papers (C) know more (+0.11 over A) (§6.13) but are just as late

Line A: from "the title helps" to "this paper, late, graded by fame"

The pilot was right that the title matters. It could not see that the target advantage is real but confined to the last layers, so the 7-layer average and the L56 endpoint both diluted it. Restricting the full cohort to the pilot's A/B groups (R0 only) changes nothing: the R1 papers do not drive the result.

Line B

What can each readout see?



small samples asked

Is J-Lens better than Vanilla? Is the fitted lens stable?

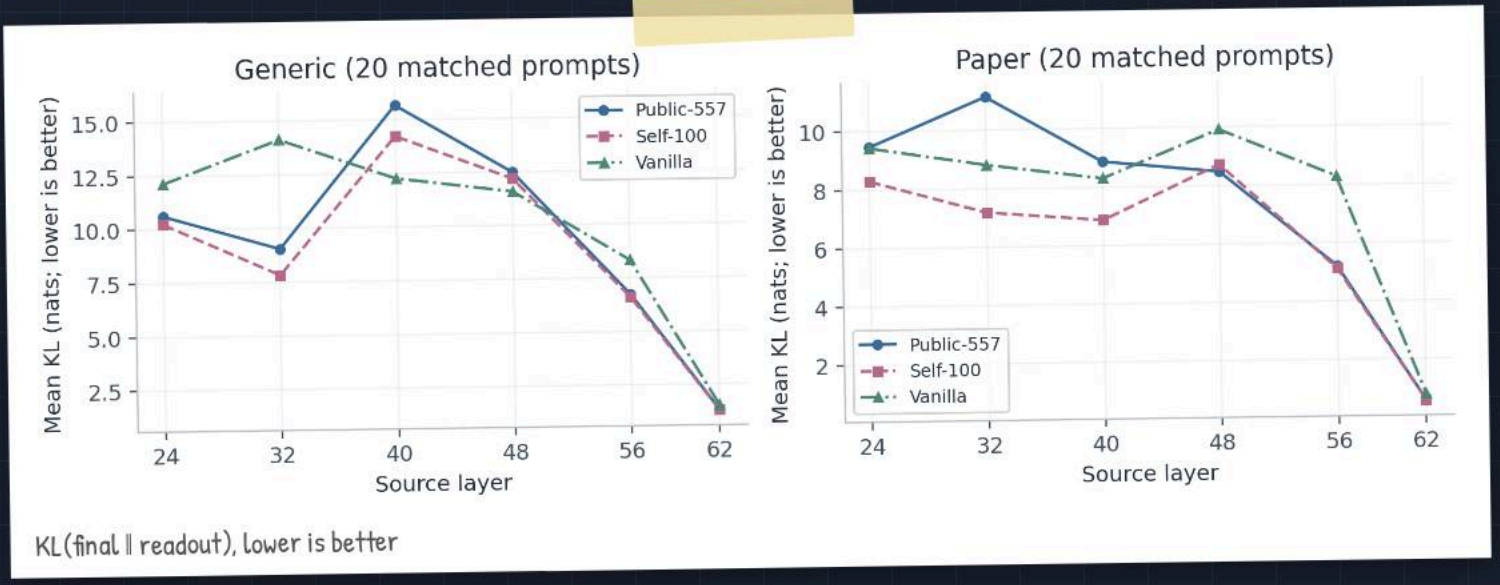
→

the full cohort answered

Lenses read name shape earlier, not paper identity; a probe at the title token reads paper-specific information

Two lenses: Public-557 vs Self-100

Can a middle layer approximate the final next-token distribution? (20 generic + 20 paper prompts)

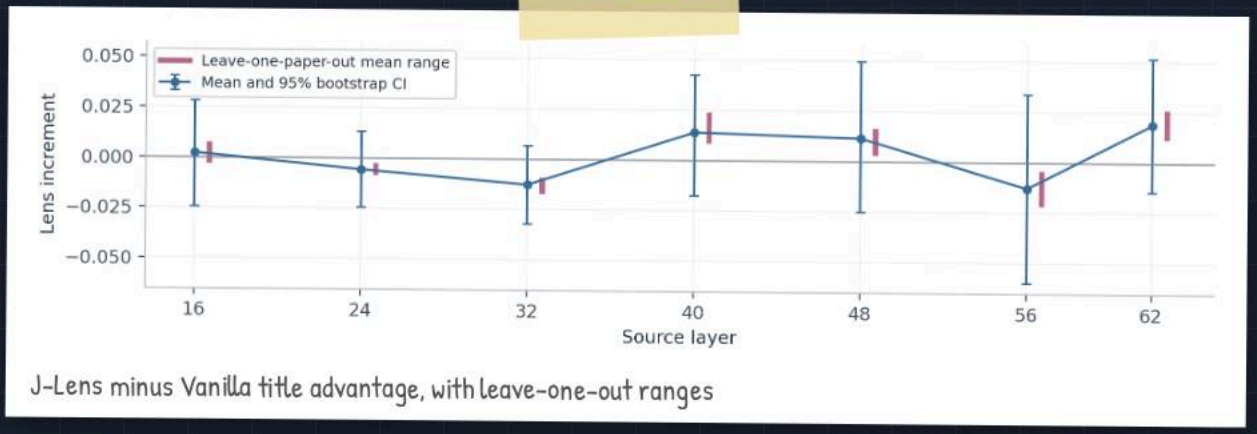


- Both checkpoints numerically valid (63/63 and 6/6 finite)(matrices.csv; frobenius_norms.csv)
- Self-100 has lower KL on 6/6 generic and 5/6 paper layers (paper L32: 11.1 → 7.2) (20 + 20 prompts; metrics.csv)
- But at paper L62, Self-100 KL beats Vanilla (0.58 vs 0.81) while Top-10 overlap is worse (0.36 vs 0.72)(metrics.csv)
- Author pilot: both lenses 14/20 full-name wins at L48(20 papers, layer_summary.csv)

Lesson: a better distribution match does not mean better author ranking. Fidelity cannot stand in for the task.

Cross-stage audit: does J-Lens add value?

Read-only joins of Stages 5–7.3; 20,000 bootstrap draws



J-Lens minus Vanilla title advantage, with leave-one-out ranges

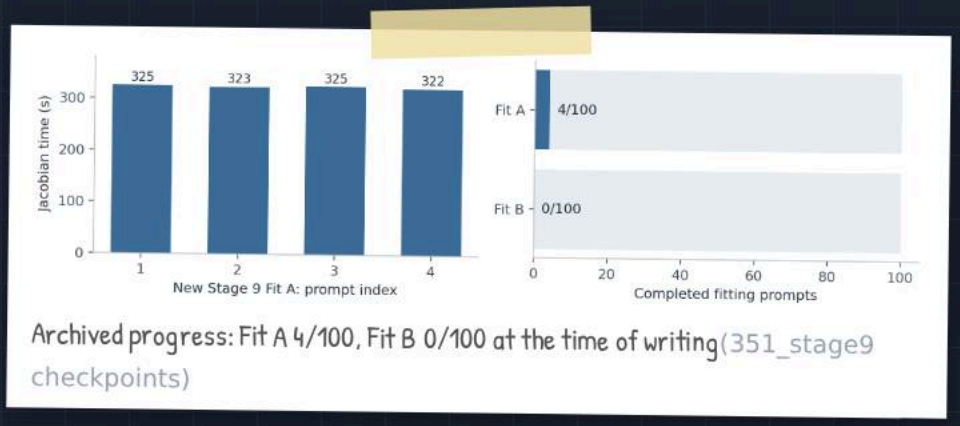
Stage 5 vs 7.3 at L56	r = 0.45, ρ = 0.47; signs agree 32/50 (50 papers, cross_stage_correlations.csv)
J-Lens − Vanilla	+0.002 [−0.008, +0.012]; median −0.002; 24/50 positive (jlens_increment_bootstrap.csv)
A−B of that increment	−0.014 [−0.033, +0.004] (20 pairs)

Dropping one paper can flip the sign of the J-Lens increment. The pilot shows **no added title-specific value** from J-Lens over Vanilla. Everything so far is exploratory: L56 was inspected repeatedly.

Stage 9: is the lens stable across fitting data?

New FullFit protocol, different from the historical 50+50 split

	Historical split-half	Stage 9 FullFit
Fit data	Self-100 split 50 + 50	two disjoint 100-prompt sets
Holdout	20 generic + 20 paper	independent 100 WikiText
Layers	6	7 (adds L16)
Status	complete	only Fit A completed; Fit B and holdout not run



Archived progress: Fit A 4/100, Fit B 0/100 at the time of writing (351_stage9 checkpoints)

- WikiText-103 train, 100–1,200 chars, deduplicated by hash, seed 3519001
- ~324 s Jacobian time per prompt, ~62 GiB CPU RAM (full_A.log; benchmark log); per-prompt checkpoints

Only Fit A was finished, and Stage 10 used it as "fitA". Without Fit B and the holdout, fitting stability is **untested**.

Layer-by-layer readout

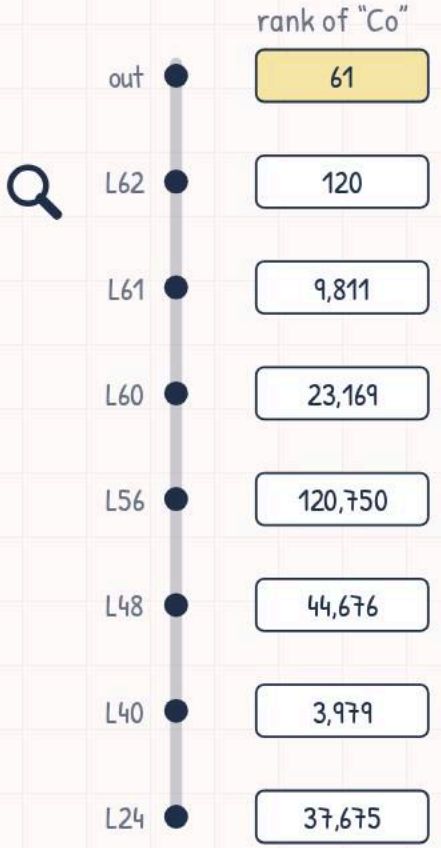
Coifman, Bonanno, Rafaeli (2007) · logit lens

1 prompt → 2 state at name → 3 lens → vocabulary → 4 rank of true token → 5 mean

nlr = log10(1 / rank)

rank = 1 + #{tokens scored above the true one}



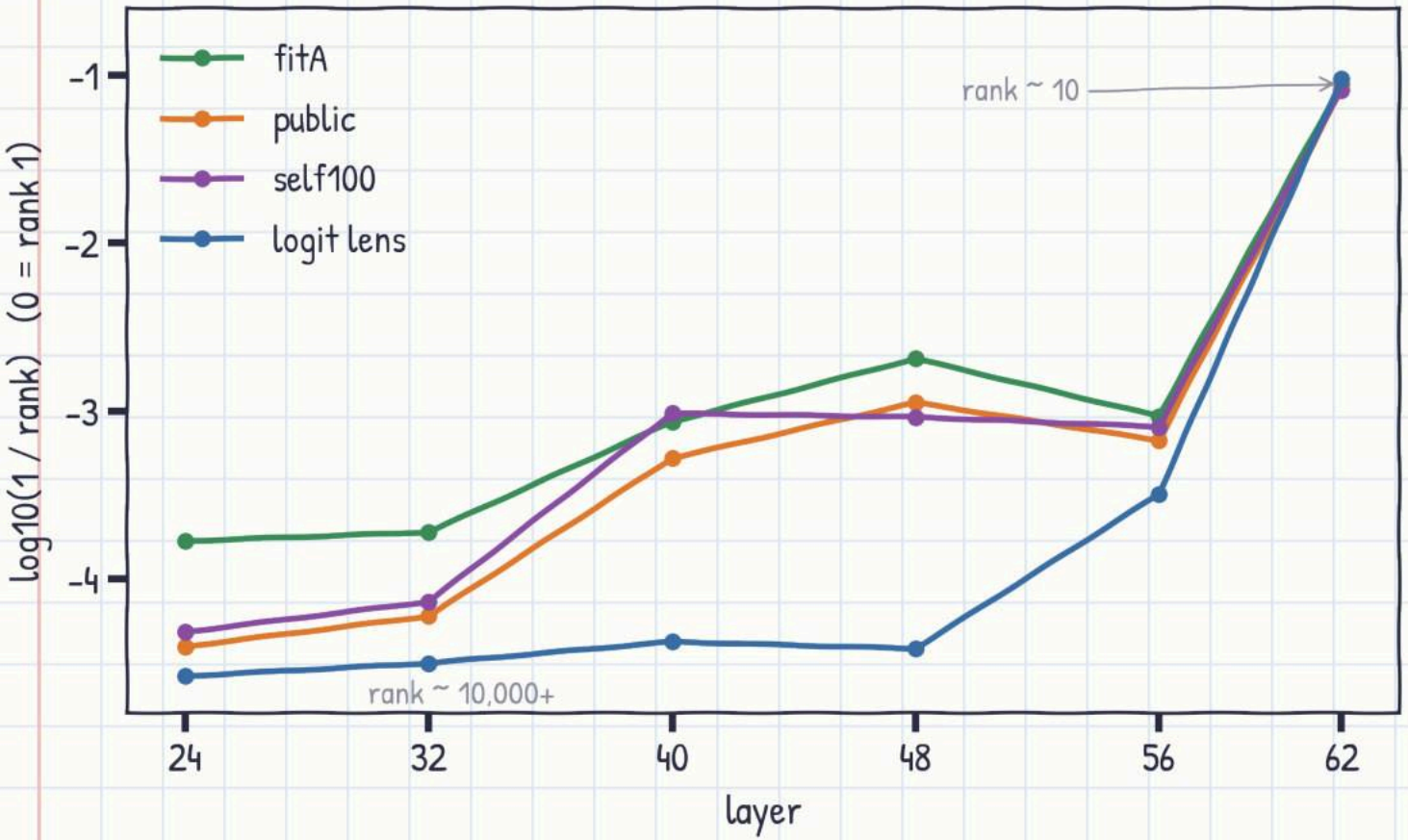


…titled | Affect | dynamics, | … | (2007)? | Authors | : | Kar | in | G | . | Co | if | man

Ranks of "Co" (author 1, token 5), true title: 351_stage10/raw/logit/L{24…62}/shard_011.npz row 25029; out = raw/final/L63

Names become readable late

all 9,108 papers · true title (Stage 10 cohort, §6.1)



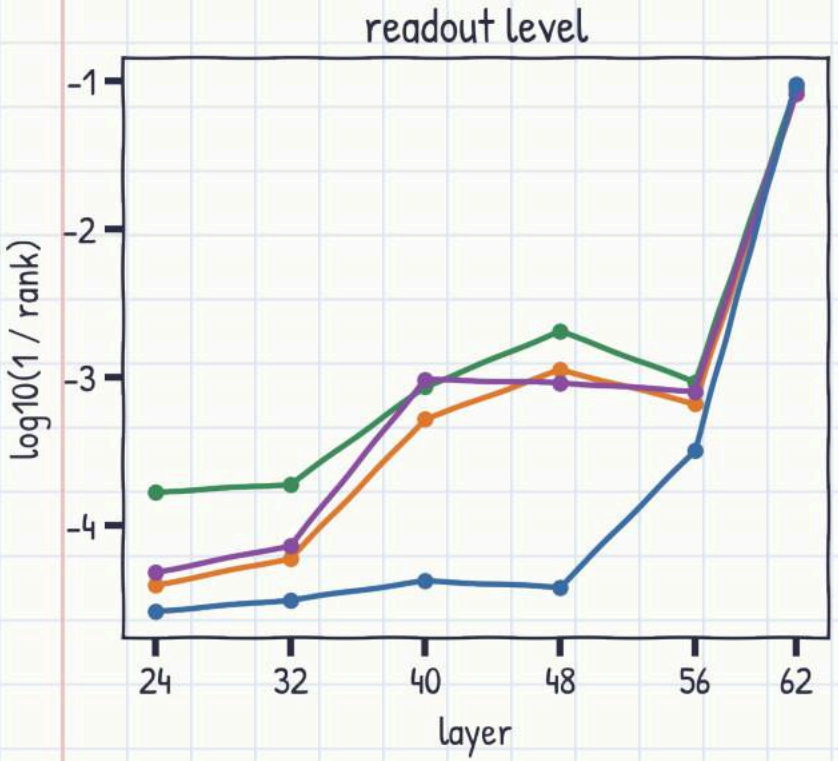
Logit lens: almost nothing until **L48**

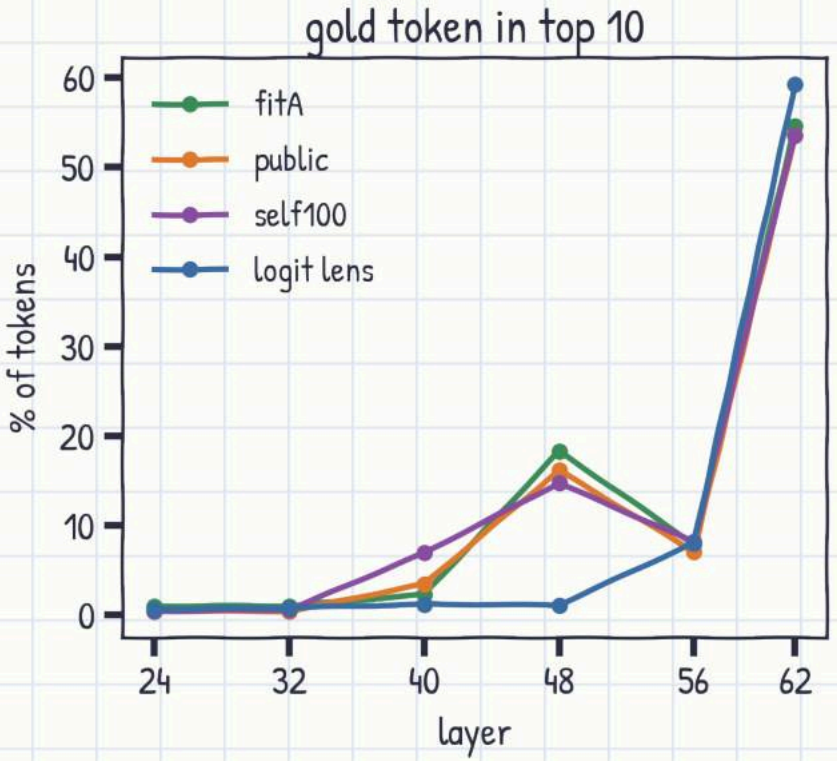
Jacobian lenses read more from **L40**

Everything jumps at **L62** (rank ~10) (Stage 10, logit L62 -0.93, §6.1)

Which lens reads best?

all papers · true title





Middle layers:

fitA best; at L48 15–18% top-10 vs 1% for logit

(Stage 10 X3, 9,108 papers, §6.9)

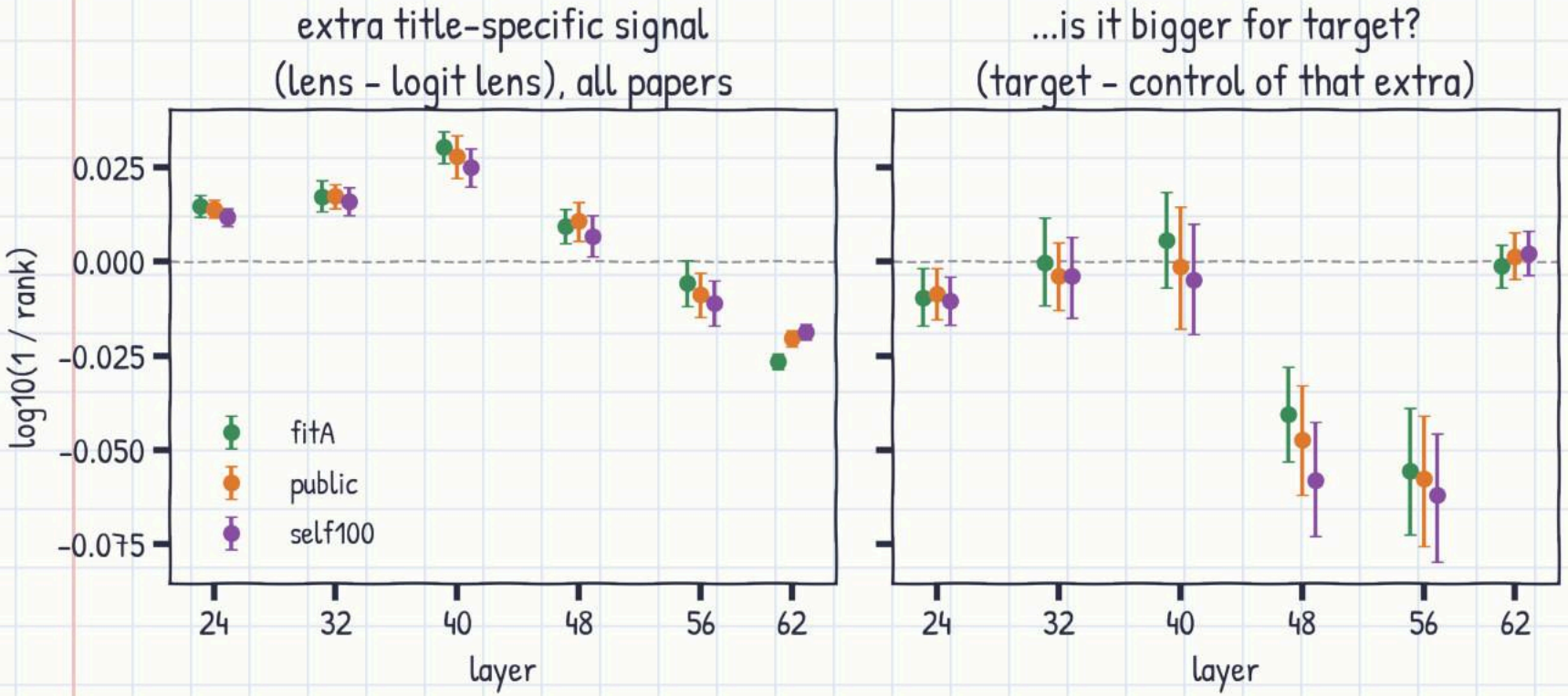
L62: logit lens best (59%)

(same, L62)

Stage 10

Do lenses reveal hidden paper knowledge?

paired differences, 95% CI



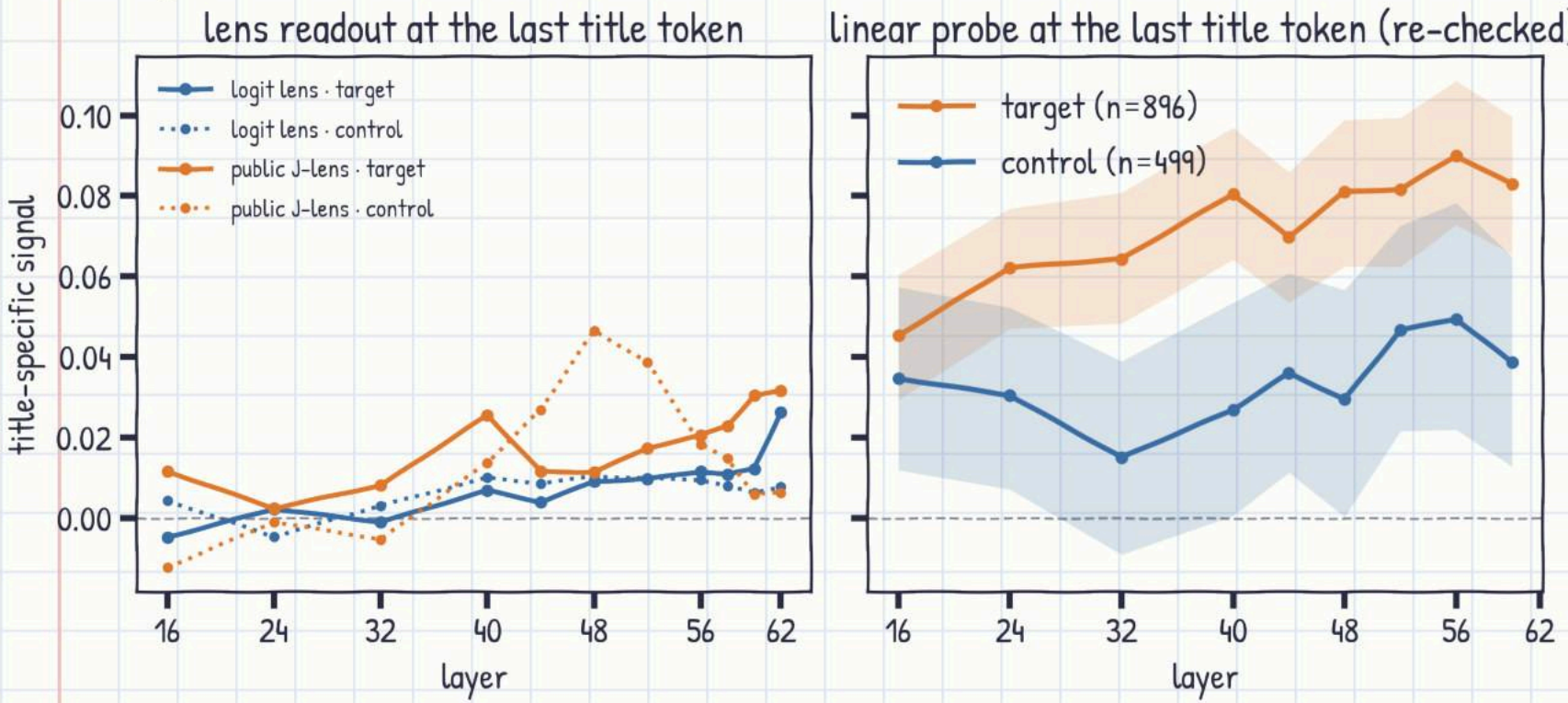
Extra signal: **+0.01 to +0.03** (Stage 10 X3, L24–L48, 9,108 papers, §6.9)

vs ~0.44 at L62 (same, L62)

Not bigger for target
(even negative at L48–56)

Read early: a probe finds it at the title

title-specific signal at the last title token · lenses: all 9,108 papers · probe: re-checked, title reuse removed: 1,863 papers (896 target, 499 control) (Stage 11 E4; Stage 13 E4b)



lenses: **~0.01–0.03** (Stage 11 E4, 4,621 + 2,220, §6.11)

probe: **0.05 -> 0.08** (Stage 13 E4b, 1,863 papers, first name token only)

target > control by **+0.04–0.08** (E4b, 896 vs 499, within bin)

re-checked: team folds, title reuse removed

scores the first name token only (mostly an initial); subtracts false_0 only

read early (title, from L24), **spoken late** (names, L57–61)

Probe scores only the first name token (mostly an initial); all-token, by-role version (addendum 9) in progress.

Stages 1–9 · 20–200 papers

What we saw

- Self-100 fits the final distribution better; author ranking not better
- J-Lens − Vanilla ≈0 (7.4: +0.002; Stage 8: −1.4 pp)(50 / 200 papers)
- Stage 9: only Fit A fitted, stability not tested

Stages 10–11 · 9,108 papers

What scale showed

- Jacobian lenses: top-10 rate 15–18% at L48 vs 1% for logit lens(9,108 papers; X3_means.csv)
- Extra title-specific signal only +0.01–0.03(L24–L48; X3_means.csv), not target-specific
- At L62 the plain logit lens is best
- Linear probe at the title token: +0.05 → +0.08 (re-checked) (E4b, scores only the first name token; e4b_trainstd_compare.md), paper-specific

Line B: the information is there, but not in a form the lenses read

Both phases agree: no lens reveals hidden paper knowledge in middle layers. Scale adds the positive side: a trained probe can decode it at the title, so the limit is the readout, not the absence of information.

Line C

When does it appear, and when is it used?



small samples asked

Which layer peaks? (L56 kept standing out)

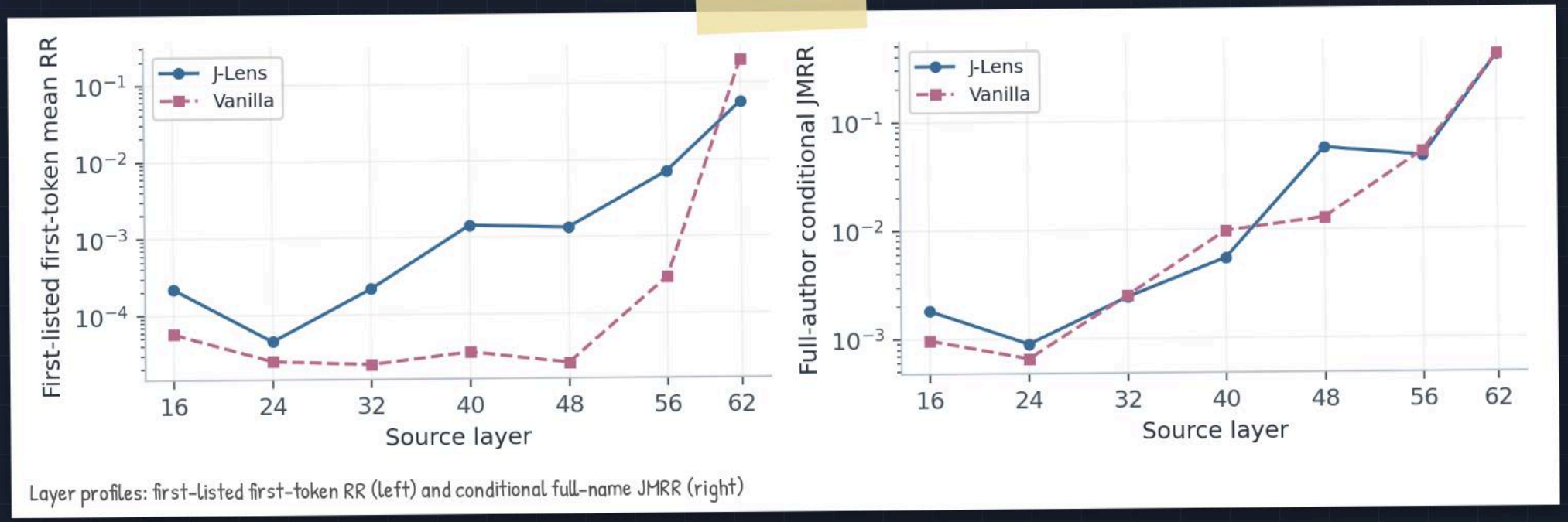
→

the full cohort answered

Readable late (L57–61) but read from the title early (L24–50)

First readout: is any author signal there?

20 papers, 93 authors (A8 / B8 / C3 / D1); ~91 s per paper, ~29.5 GiB GPU(351_stage4_readout.log)



Layer profiles: first-listed first-token RR (left) and conditional full-name JMRR (right)

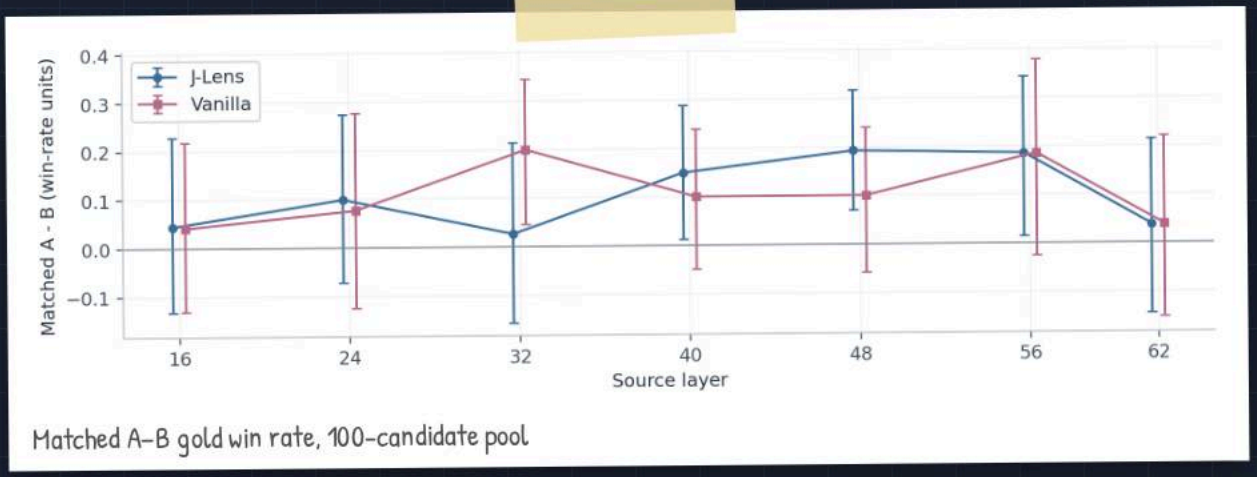
- The pipeline works and J-Lens rises above Vanilla in middle layers
- L56 first-author mean RR 0.0070, median only 0.0009(20 papers; audit_report.md table 1)

One paper = 71% of the L56 mean

Dropping it: 0.0070 → 0.0021.(same file) A layer peak can be a single easy paper, so every later stage adds controls.

Gold vs 100 real false authors

50 matched (same field, token count, year) + 50 random names per paper; 75,000 scored pairs(candidate_pairwise_scores.csv)



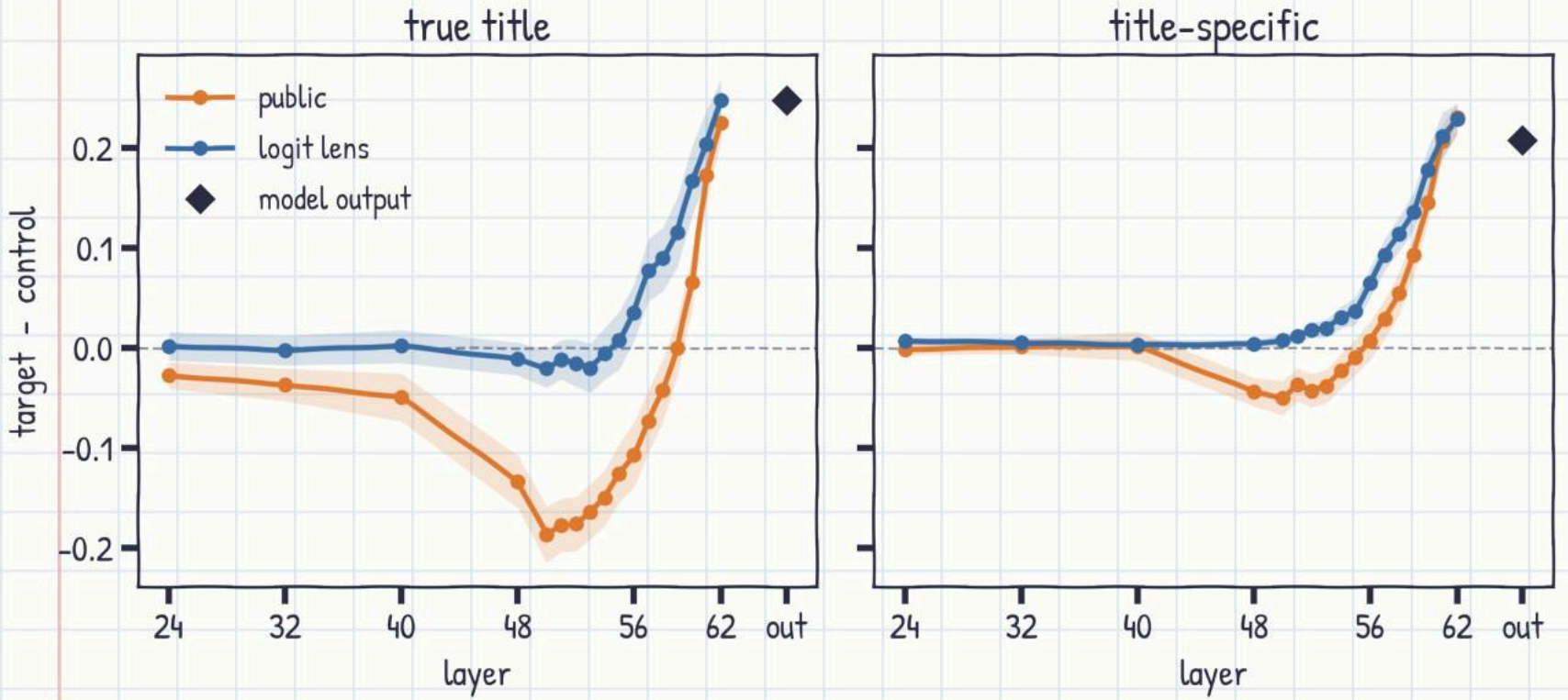
Matched A–B gold win rate, 100-candidate pool

- J-Lens A–B: L40 +0.15, L48 +0.19, L56 +0.19 (CIs above 0); L62 +0.04 (spans 0)(20 pairs, matched_ab_bootstrap.csv)
- 7-layer integrated: +0.105 [−0.012, +0.229](20 pairs; recomputed 10-30)
- 156/5,000 false names share the gold first token (ties) (quality.json)
- 100 names → only ~46 distinct first tokens(mean of 50 papers; candidates csv)

Gold beats real names in middle layers for A more than B. But the same names might win under any title: 7.3 tests that.

Target pulls ahead only in the last layers

target − control, within citation bin, 95% CI



Middle layers: **no head start**

L62 / output:

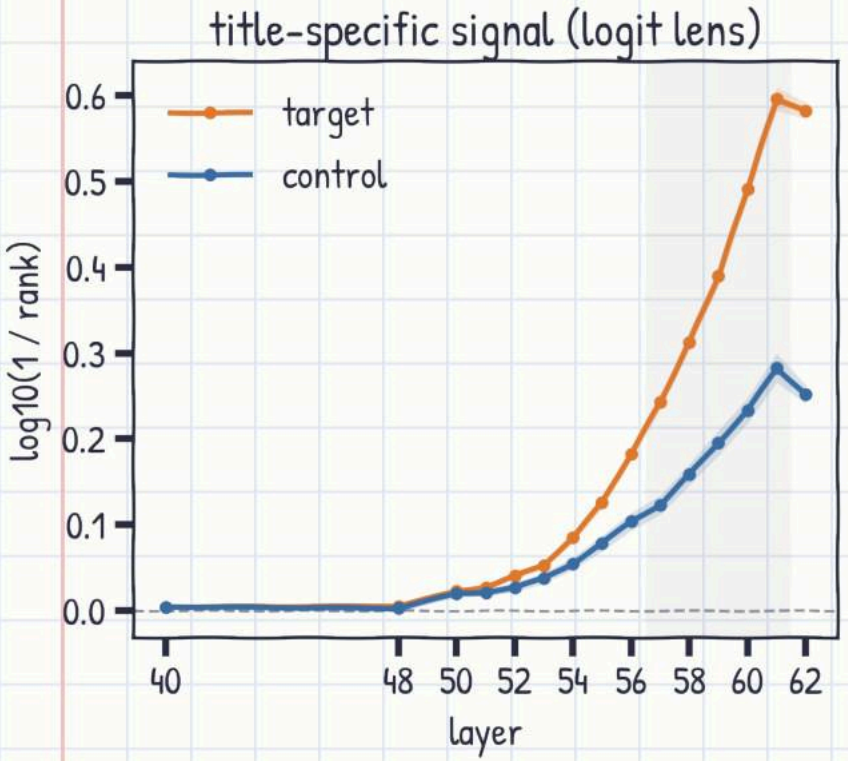
+0.21 to +0.23 (Stage 10, L62 + final L63, 3,734 + 2,182 papers, §6.4)

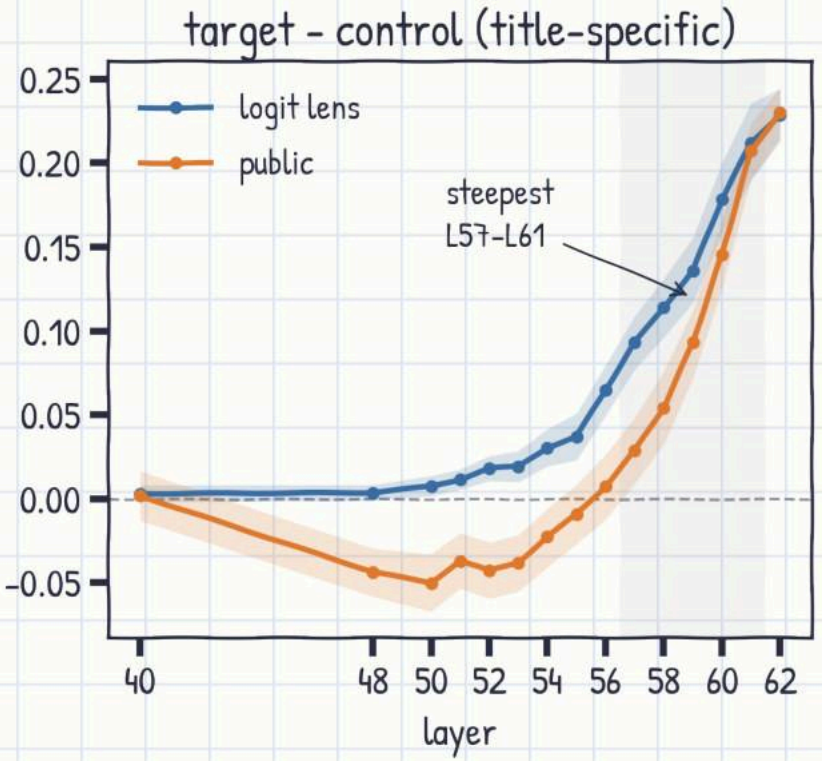
almost all title-specific

R0 only (no author written): output +0.199 vs +0.207 (Stage 10, 3,423 + 2,151 R0 papers, §6.13)

It builds up over the last dozen layers

fine layers L40–L62 · title-specific signal · 95% CI





Logit lens: target - control

+0.007 at L50

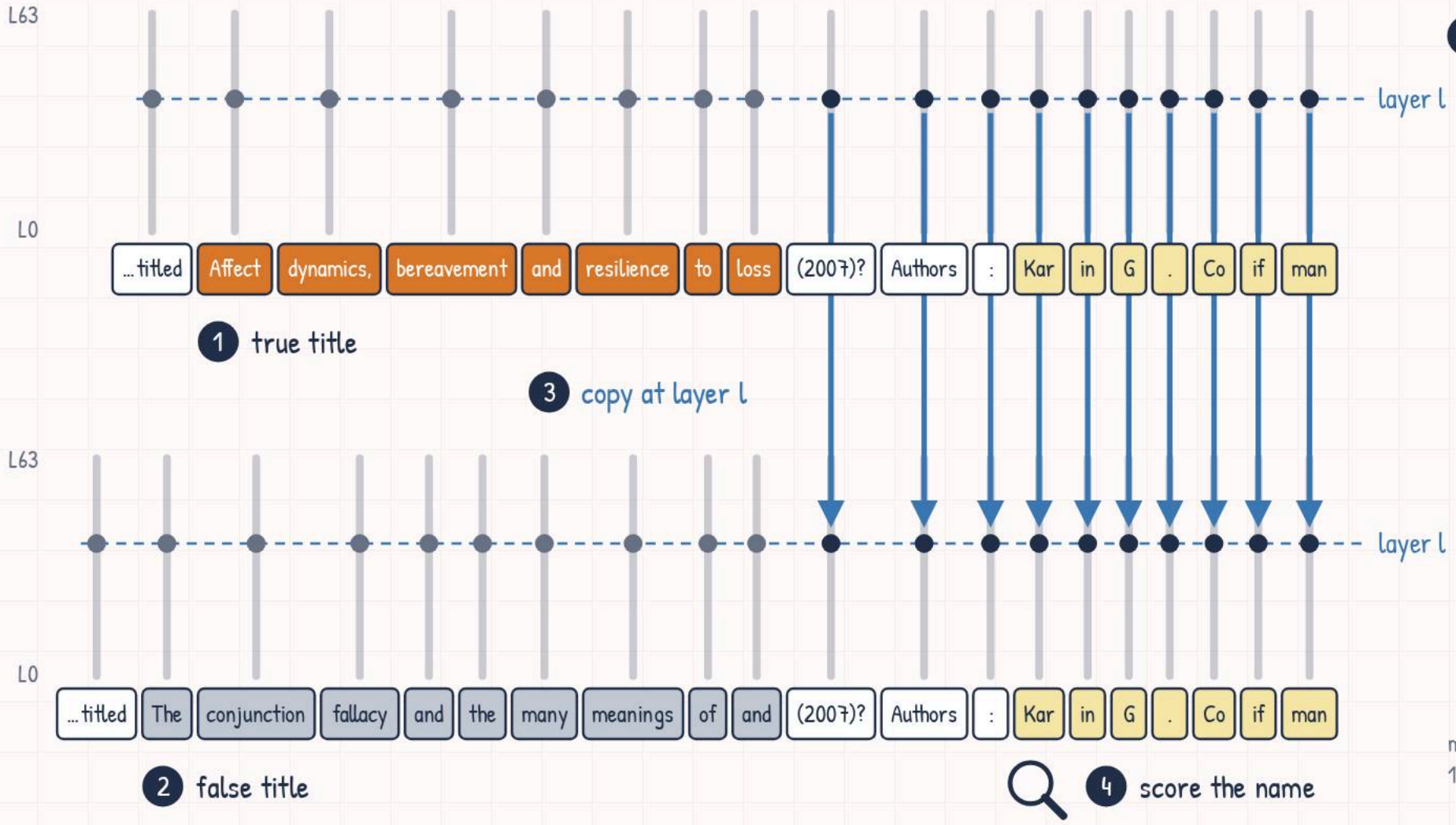
+0.23 at L62 (Stage 10 logit lens, 3,734 + 2,182 papers, §6.10)

steepest rise **L57–L61**

Replacement (patching)

Coifman, Bonanno, Rafaeli (2007)

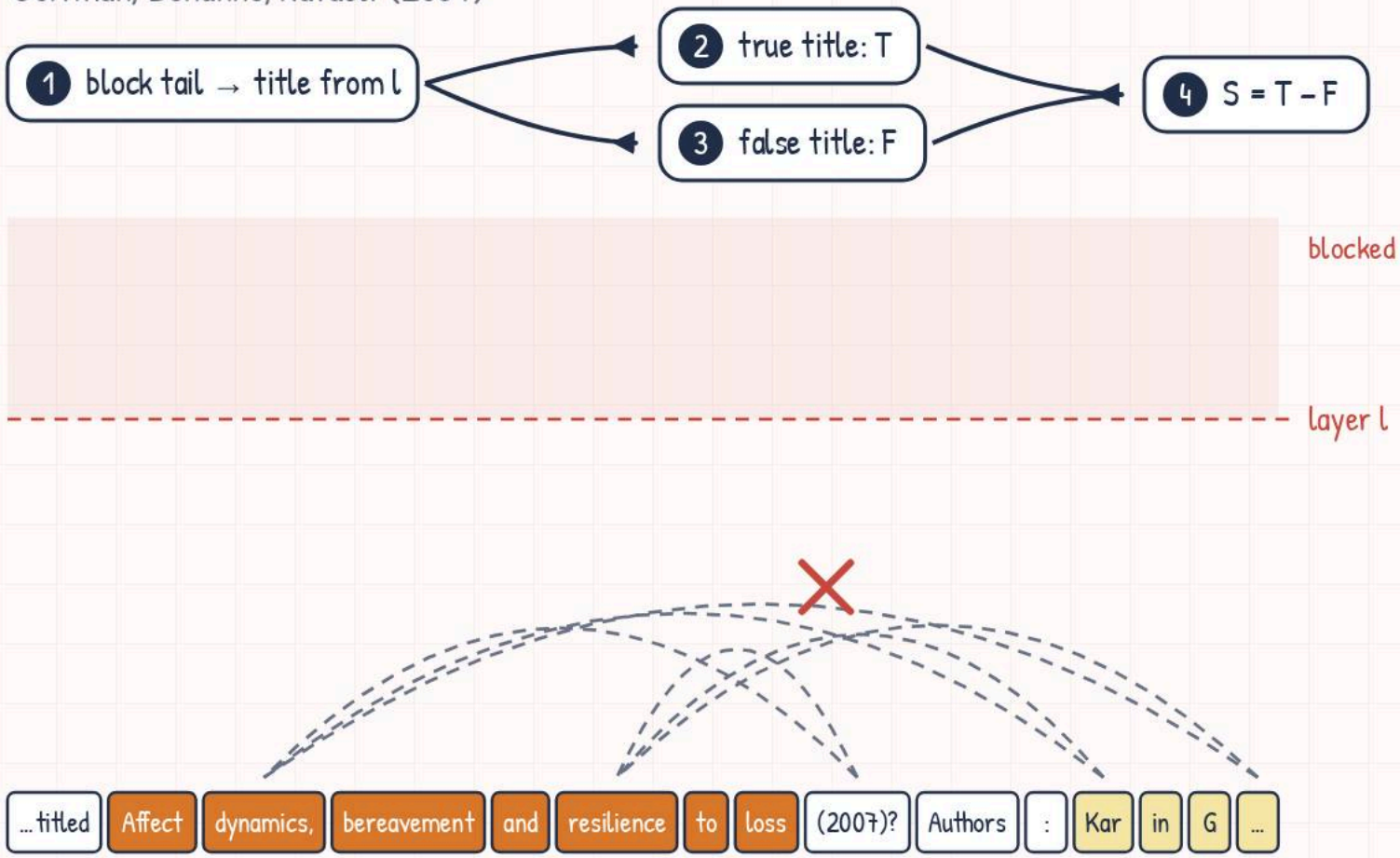
$$R(l) = (\text{patched} - \text{false}) \div (\text{true} - \text{false})$$

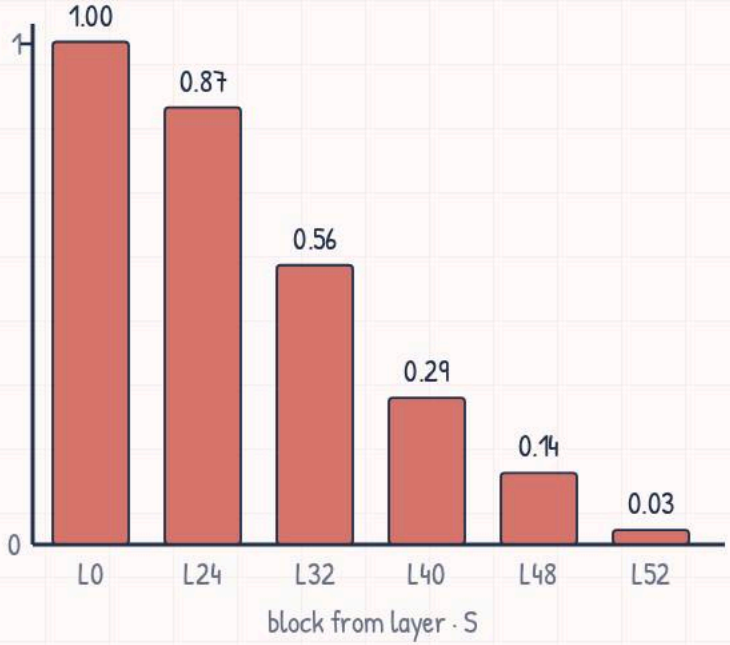


R(l): Stage 13 X6, mean of 5 false titles, all 1,186 papers; 381_stage13/summary_x6/x6_recovery.csv

Knockout (block attention)

Coifman, Bonanno, Rafaeli (2007)





L52: target 0.018 · control 0.051

S(l) = T(l) − F(l)

loss as share of (true − false), 1,186 papers

Stage 11: intervene, do not just read

593 + 593 papers: min(60, n_t, n_c) per citation bin, seed 11 (Stage 11 sample.parquet, §6.11)

patch uses false_0 · knockout masks all heads · all 9,108 for probe and chain (Stage 10 manifest.json)

patch

copy the true-title hidden state into the false-title run at one layer

knock out

block attention to the title tokens (all heads) from layer l to the last layer

probe

linear map from the last title token to the output state

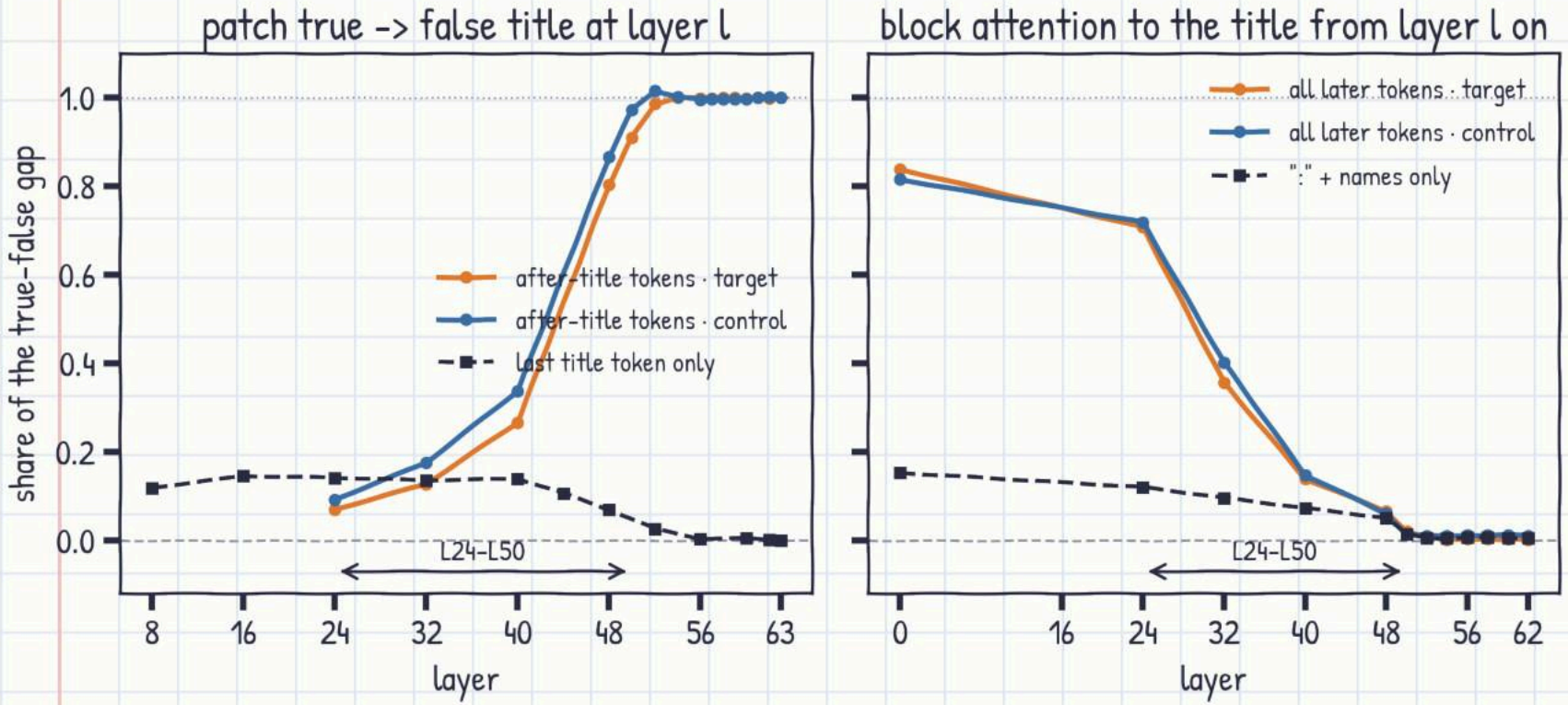
chain

"Authors: A1, A2, ..." instead of each author alone

Score: rank of the correct name token at the model output; "share of the true-false gap" = (intervened - false) / (true - false).

The title is read in L24-L50, early

model output · share of the true-false gap · CIs about +/-0.02 (Stage 11 E1/E2, 593 + 593 papers, §6.11)



L52+: title no longer needed

the information enters the **name positions at L44–52**, partly via the tokens right after the title

same timing for target and control

Stages 1–9 · 20–200 papers

What we saw

- Stage 4 peak at L56 driven by one paper (71%)(audit_report.md)
- 7.2: A–B win-rate positive at L40–L56
- 7.3, 8: L56 title advantage; L56 became the primary layer

Stages 10–11 · 9,108 papers

What scale showed

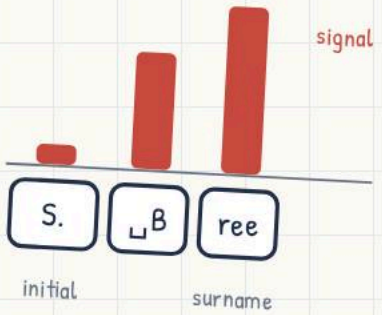
- L56 is where it *starts*; steepest rise L57–L61
- Patching: copying tail states recovers 80–87% at L48, 100% from L52(Stage 11 E1, 593 + 593; x6_report.md)
- Knockout: blocking the title from L52 on costs nothing

Line C: "read early, spoken late"

L56 was a real landmark, but not the peak. Interventions add what readouts cannot: the title is consumed in L24–50, and the answer only becomes readable at the name a dozen layers later.

Line D

Which token carries it?



small samples asked

First token or the rest of the name?

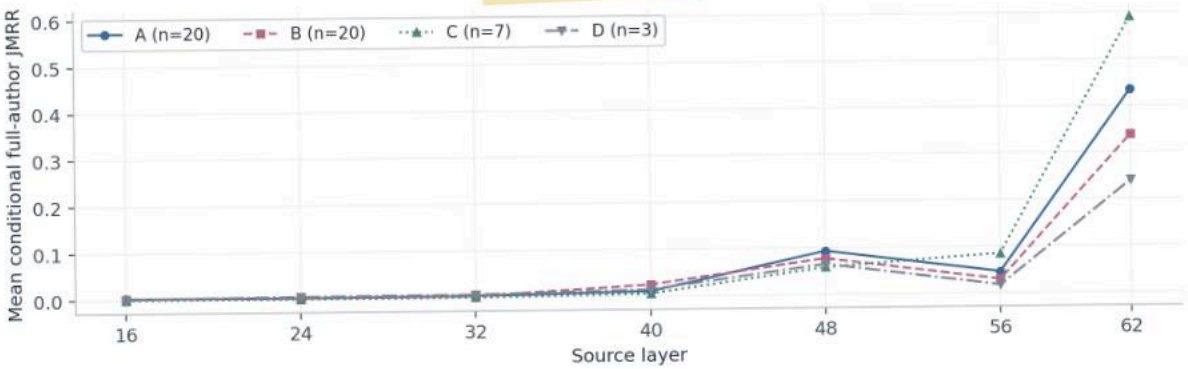
→

the full cohort answered

Depends on name format; the information enters the name positions at L44–52, partly via the tokens right after the title; co-authors partly replace the title

From first token to the full name

50 papers, 227 authors, 972 name tokens(token_layer_scores.csv); every listed author scored with teacher forcing



Conditional full-author JMRR by group

Group	Integrated JMRR(integrated_group.csv)
A (20)	0.085 [0.075, 0.096]
B (20)	0.068 [0.057, 0.080]
C (7)	0.106
D (3)	0.050

L62 A–B +0.097 [0.028, 0.163]; Vanilla also +0.091(20 pairs, matched_ab_bootstrap.csv)

Positive, but each token after the first is given the correct prefix. Next: split first token from the rest (7.1).

The A–B gap sits on the rest of the name

CPU-only re-analysis of Stage 6; 10,000 pair-bootstrap draws



Integrated A–B JMRR by token position

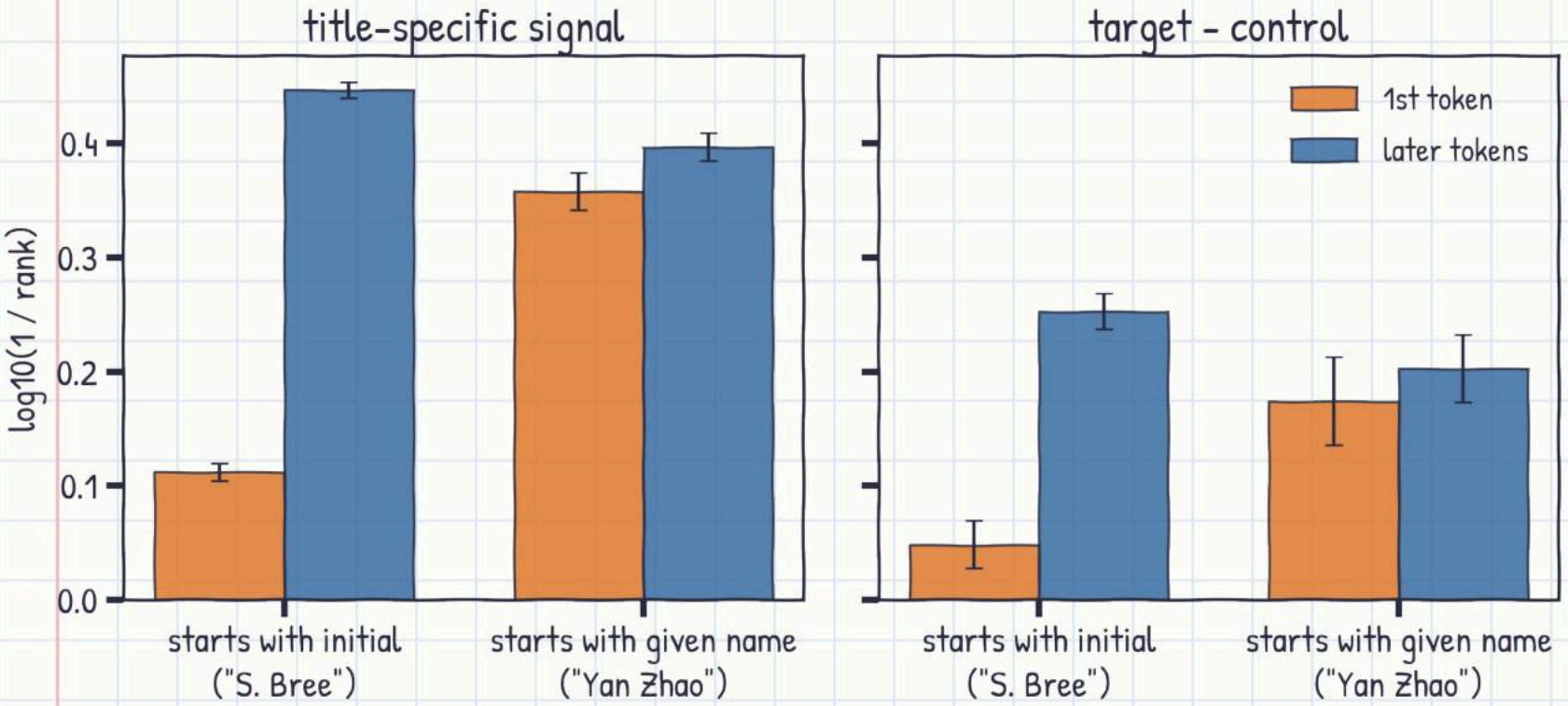
Endpoint	A–B (7 layers) (20 pairs; layer_and_integrated_effects.csv)
First token	+0.001 [−0.011, +0.015]
Continuation	+0.021 [+0.002, +0.039]
All tokens	+0.017 [+0.003, +0.030]

At L62: first +0.014 (12/20 pairs) vs continuation +0.121 (15/20) (layer_position_effects.csv)

Completing a name is easier than starting it. A continuation gap cannot show unaided recall. **Open:** is this about initials vs surnames?

First token or the rest of the name?

model output · split by name format

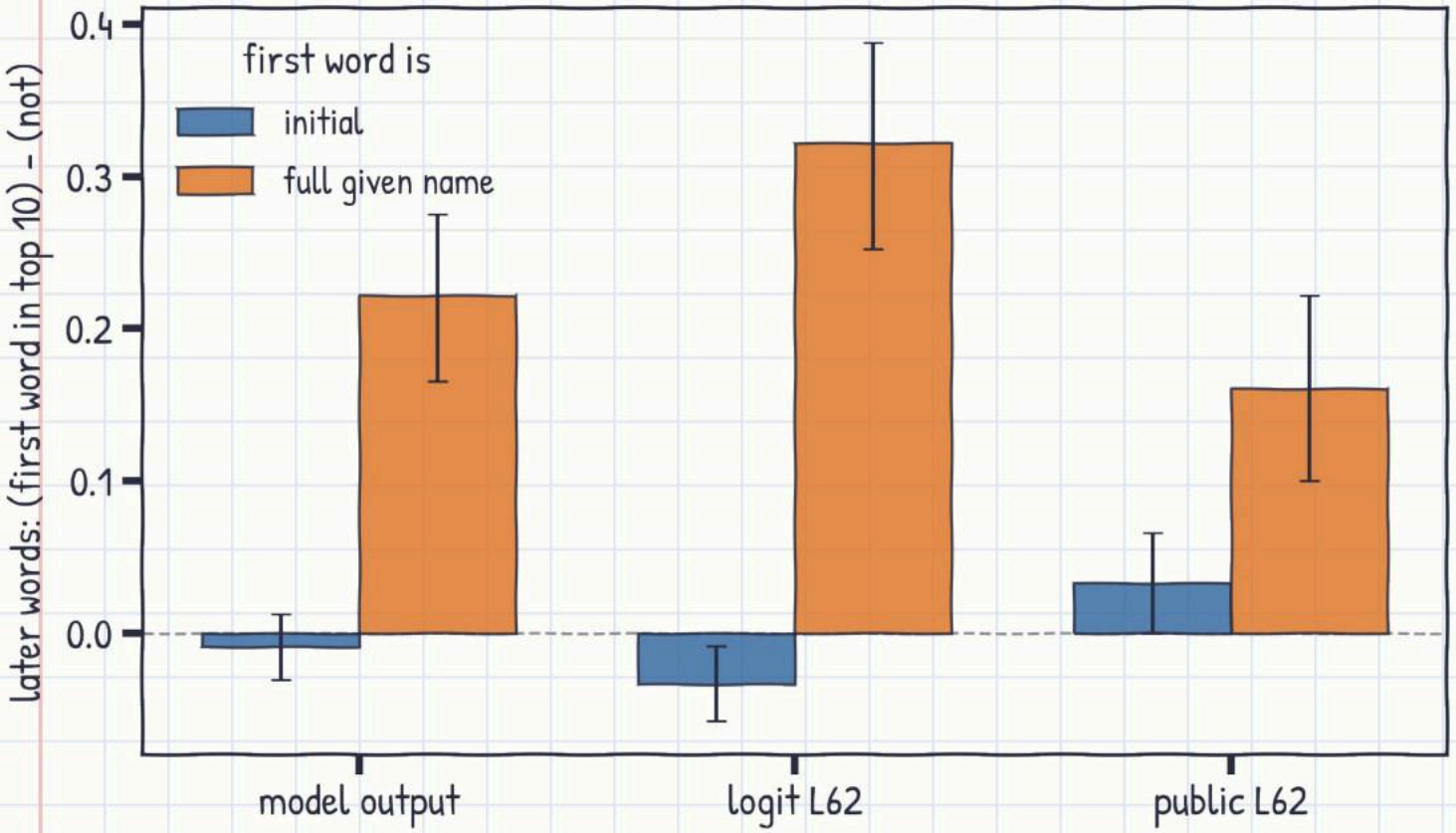


Initial: carries little; signal is on the surname

Full given name: carries about as much as the surname

Knowing the given name helps with the surname

true title · later words when the first word is / is not in the top 10



Given name in top 10

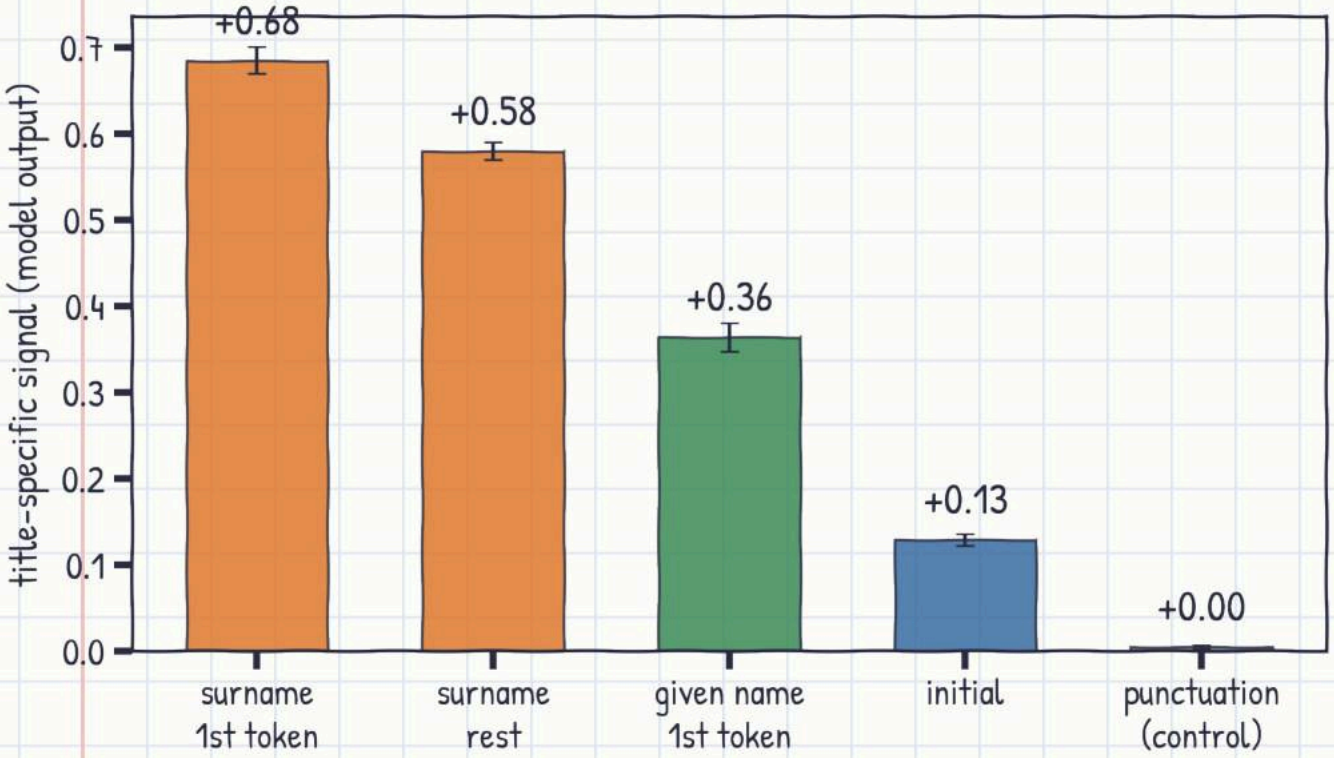
surname +0.22 (output) (Stage 10, 8,876 authors, §6.7)

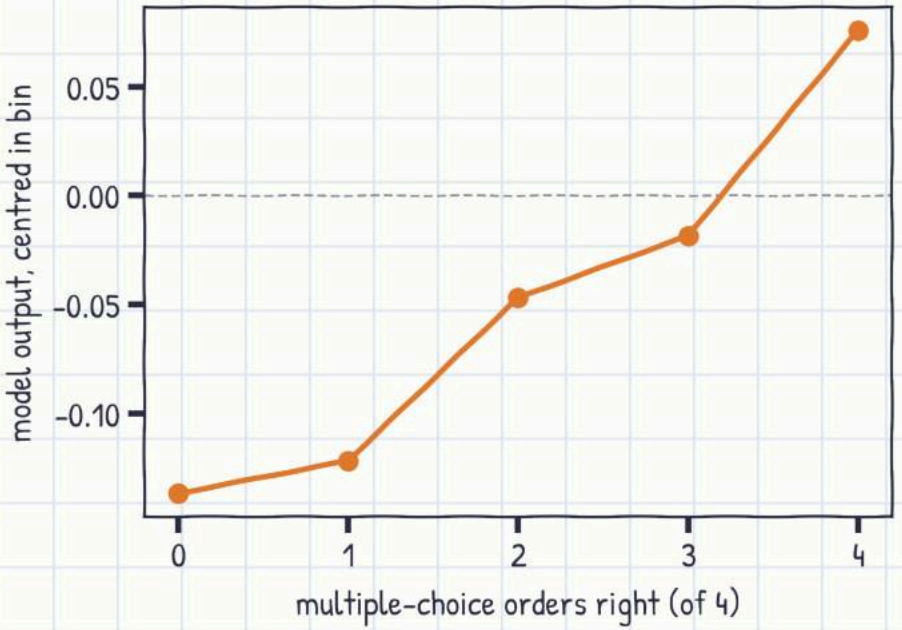
+0.32 (logit L62) (same)

Initial in top 10: no help

Where the signal sits, and a dose-response

model output

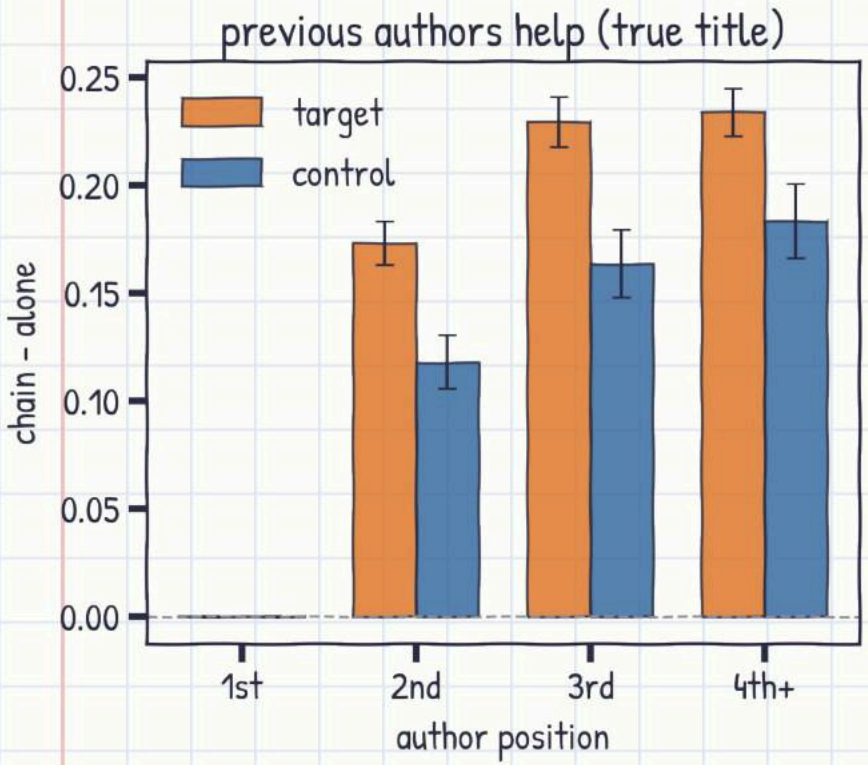


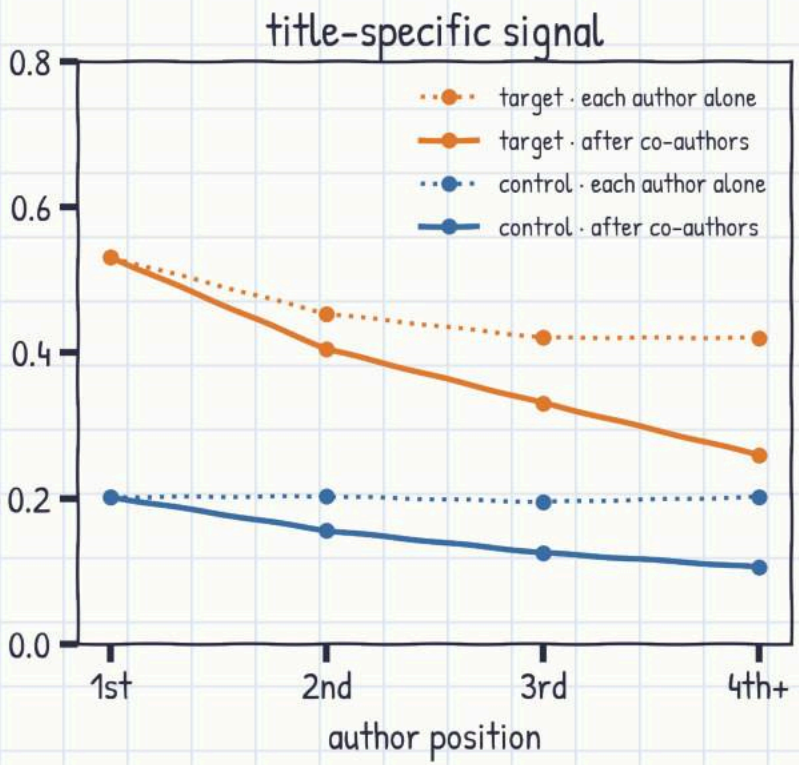


Right: R0/R1 papers, centred within citation bin; Spearman rho = 0.27, $p < 0.001$. (Stage 10, 8,894 papers, §6.6)

Co-authors help, and partly replace the title

model output · all 9,108 papers · 95% CI (Stage 11 E3, §6.11)





later authors **+0.17 to +0.23** (Stage 11 E3, 4,621 target, §6.11)

title-specific part **shrinks**

target > control still

Stages 1–9 · 20–200 papers

What we saw

- A–B lives on continuation tokens (+0.021), not the first token (+0.001)(7.1, 20 pairs)
- Open question: is that teacher forcing, initials or surnames?

Stages 10–11 · 9,108 papers

What scale showed

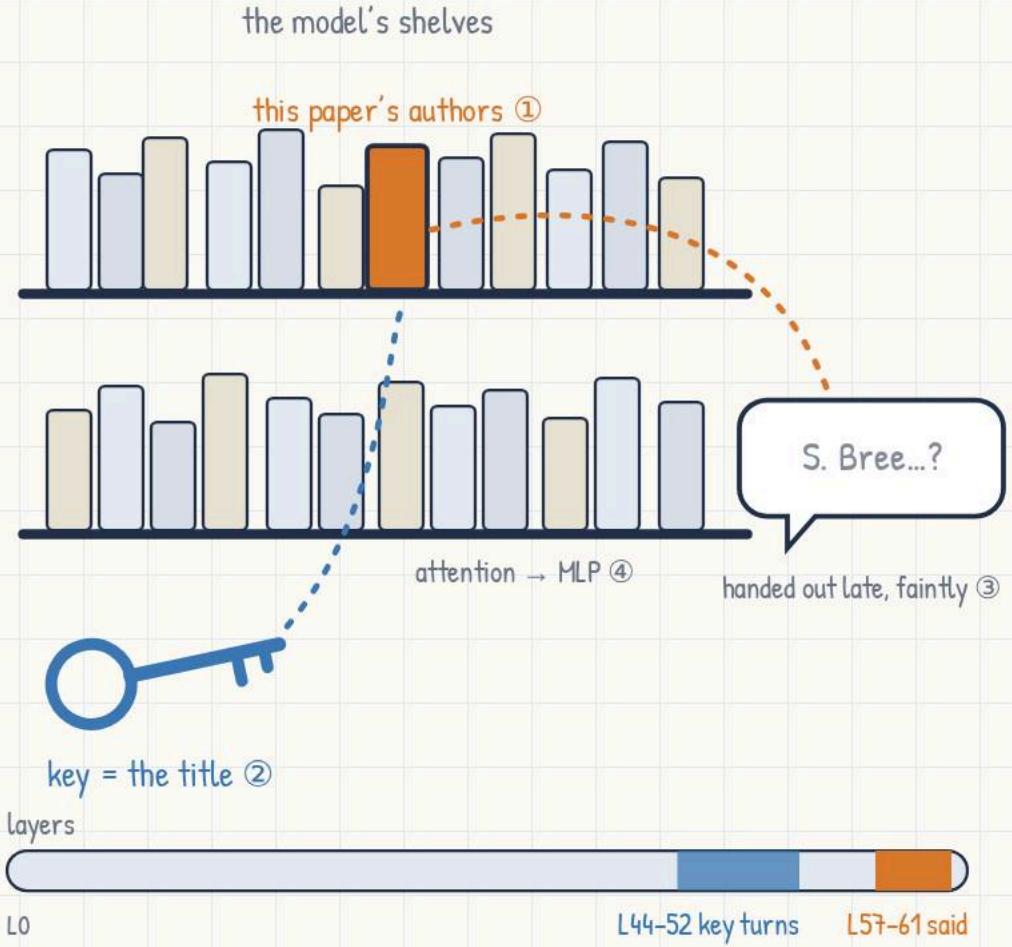
- 73% of names start with an initial(24,739 / 33,780; §6.7), which carries little
- Full given names carry as much as surnames and help predict them (+0.22)(final L63; §6.7)
- Information travels title → name positions (L44–52), partly via tail tokens
- Earlier co-authors make later ones easier, title matters less

Line D: the puzzle from 7.1 is explained by name format

The pilot's "continuation" result was mostly a format effect: for "S. Bree" the surname is a continuation token. Stage 12 P1 (done) patched single positions: the name positions carry the signal (+0.85 at L52), the colon alone little (+0.17)(P1, 1,186 papers; P1_patch_pos.csv).

Keys and shelves

Is the book on the shelf, or is the key failing to get it out?



1 Is the book on the shelf? Yes

True title beats false titles (≈0.44 at L62(§6.9)); a probe on the title token reads paper-specific authors from L24 (re-checked)

Lines A, B · Stage 5, 7.3, 8D, 10, 11

2 Key = the title, read into the name positions at L44–52

Whole prompt: 0.82 at L48, ≈1.0 from L52; name positions alone 0.61 → 0.85; ':' alone ≤ 0.17(E1, P1; 1,186 papers); knockout after L52 costs almost nothing

Line C · Stage 11, 12 P1

3 Book there but stuck? Not supported

Target – control ≈ 0 at L24–L40, slightly negative at L48 (–0.04 to –0.05), grows only at L57–61 (+0.21 at output(§6.4)): retrieval itself is weak

Lines A, C · Stage 7.2, 10

4 Which shelf? Both, in order

Attention brings the title in at L48 (0.23); MLP takes over at L52–56 (0.18, 0.22)(P2, 1,186 papers). One-layer patching is a lower bound; whether the MLP step is memory lookup is not yet separable

Stage 12 P2

Caveat: a readout not finding it ≠ absent. Stage 13: ':' and name positions hold the task; the title position holds paper attributes and meaning beyond the words, not shown to be paper identity.

What the small samples taught us to watch for

- **Single-paper leverage**: one paper = 71% of the Stage 4 L56 mean (20 papers)
- **Tail-driven contrasts**: Stage 8 A–B disappears after removing 10 of 100 pairs (8D, matched_pair_differences.csv)
- **Forking paths**: L56 inspected repeatedly, then made primary
- **Candidate artefacts**: 100 false names share ~46 first tokens; 3% tie with gold (7.2 quality.json: 156/5,000)
- **Label noise**: strict "pass" 25.5% in the 0–2 citation bin (label_README.md)
- **Partial coverage**: first-listed author stands in for the team
- **Metric choice**: KL fidelity and author ranking disagree
- **Protocol drift**: primary endpoint changed between README and run

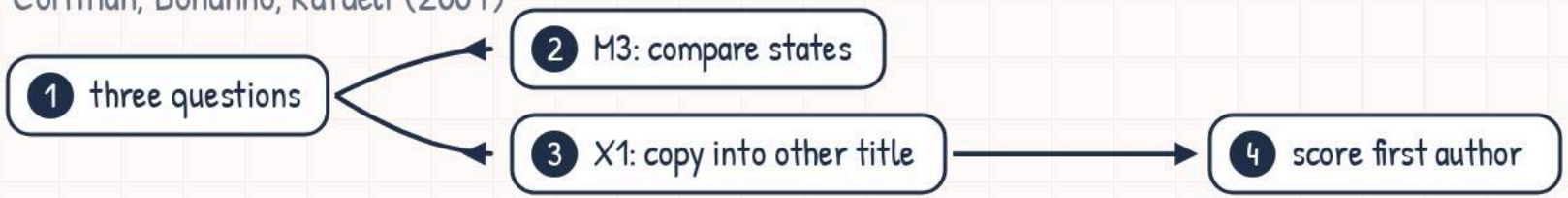
Stage 10 answered each one by design: full cohort, every author and token, all layers reported, full-vocabulary ranks, within-bin comparison, frozen inputs with parity checks. What it cannot fix: groups still come from one labelling run of the same model.

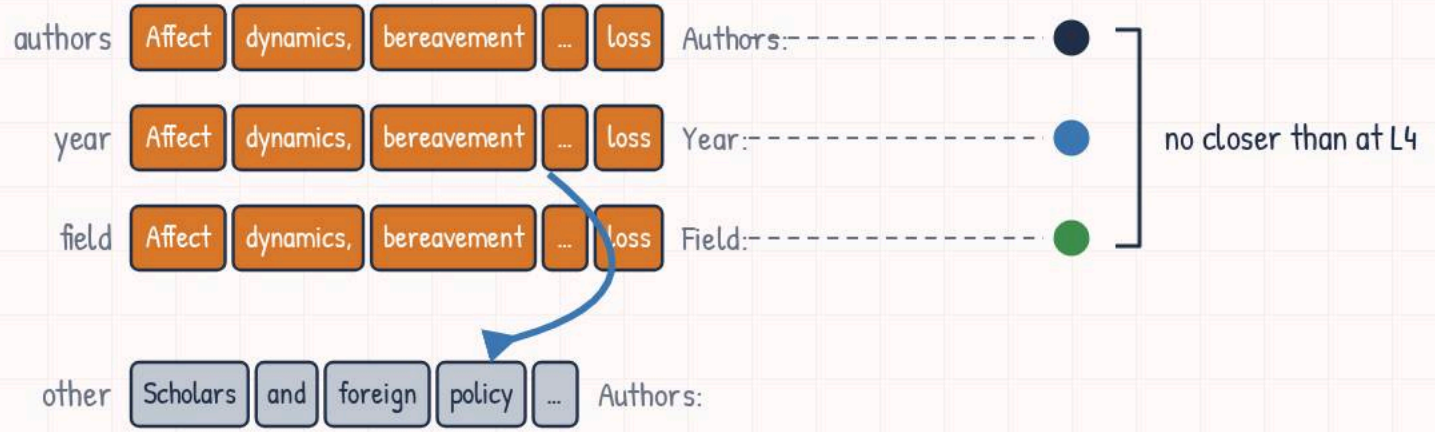
Small vs large, in one table

Finding	Stages 1–9 (20–200 papers)	Stages 10–11 (9,108 papers)	Verdict
True title helps	+9.3 pp pilot, +5.2 pp new 200 (7.3; 8D RQ1)	≈0.44 title-specific at L62 (X3_means.csv)	confirmed
Recognised > not recognised	uncertain, tail-driven	+0.21 at output (§6.4), ≈0 at L24–L40, slightly negative at L48 (−0.04 to −0.05)	confirmed, late only
Same groups (A vs B, R0 only)	A − B as above	+0.20 at output (§6.13), ≈0 at L24–L40, slightly negative at L48; R1 adds strength, not timing	comparable, unchanged
Key layer	L56	starts L56, rises L57–61; read L24–50	refined
J-Lens over Vanilla	no increment	more name shape, no paper identity	consistent
First vs rest of name	gap on continuation	explained by initials vs given names	explained
Citations	low–high inconclusive	monotonic, $\rho = 0.45$ (X2_spearman.csv)	resolved
Lens stability	split-half one split; Stage 9 only Fit A	three Jacobian lenses fitted on different text agree; fit stability rests on Stage 9 Fit A only	only one piece of evidence (Stage 9 Fit A); Fit B not completed

Changing the question

Coifman, Bonanno, Rafaeli (2007)





R from	authors	year	field
L24	**0.99**	**0.98**	**0.99**
L40	**0.92**	**0.92**	**0.92**
L48	**0.42**	**0.41**	**0.42**

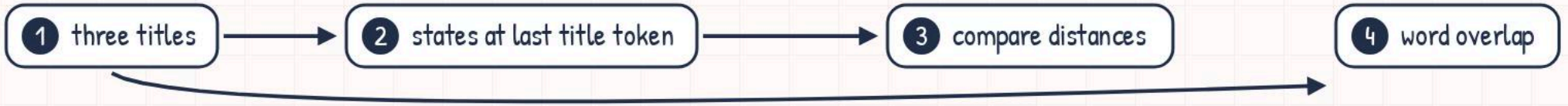
R(l) = (patched − other) ÷ (own − other)

title states copied at layer l; 582 test papers

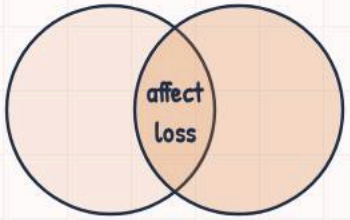
R: Stage 13 X1, test set 582 papers; 381_stage13/reports/x1_report.md

Paraphrased title (X3)

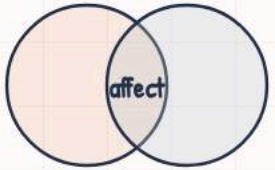
same paper in other words vs other paper, same words







para 0.22



D 0.14

G′ = cos(orig, para) − cos(orig, D)

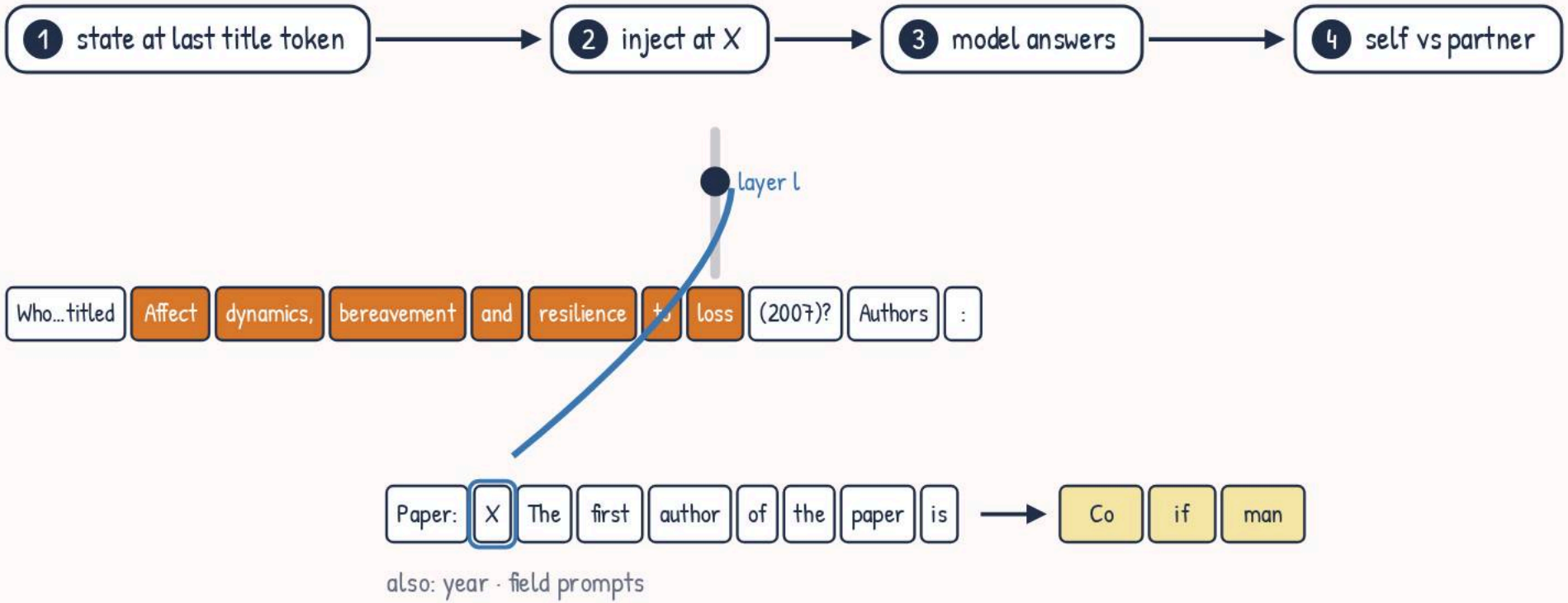
+0.32 at L8 · peak +0.46 at L36 (test set)

medians: 0.61 vs 0.15

first-author score, paraphrase − D: +0.239

Patchscope (X4)

Coifman, Bonanno, Rafaeli (2007)



D = score(own state) − score(partner state)

score = log10(1/rank) of the true answer tokens

first author from L8: +0.046

year from L40: +0.013

D: Stage 13 X4, test set, team CI; first author L8 +0.046 [0.035, 0.059], year L40 +0.013 [0.008, 0.018]; 381_stage13/reports/x4_report.md

Stage 12–13: what the middle layers hold

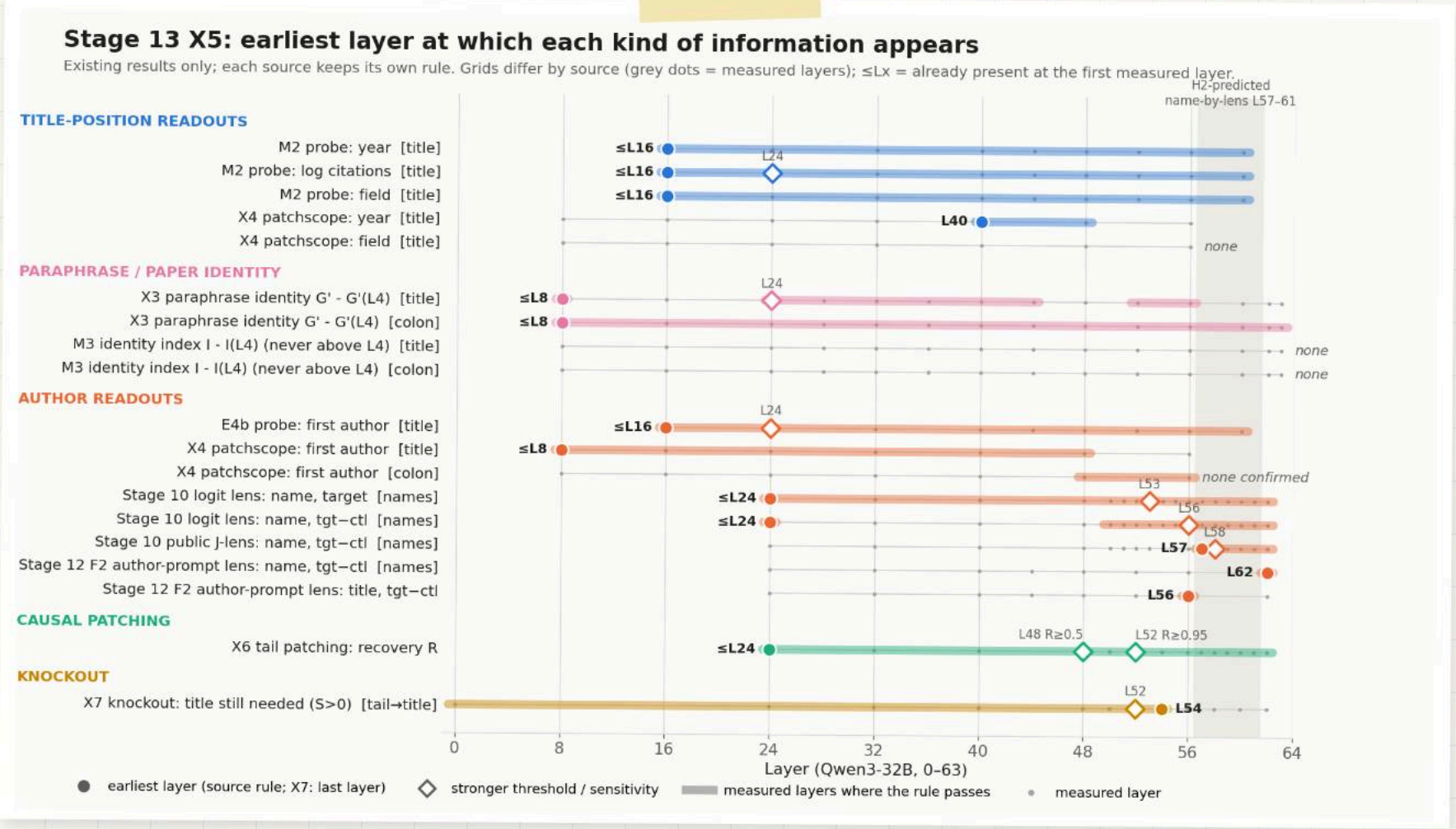
Paper facts are readable at the title early (M2) · H2 (paper identity) partly supported but not established; H3 (hidden author knowledge): insufficient evidence

假设	M1：lens 在中间层排第一的词	M2：probe 读非名字的属性	M3：换任务对照	目前结果（2026-11-01）
H1 任务 / 格式：中间层在表示“要输出作者名单”	多为 “author”、标点、名字常见开头，与标题无关	论文属性读不出或很弱	同一标题在不同问题下，中间层表示早早分开	见上文 M3、X1 结果
H2 论文身份：中间层在表示“是哪一篇论文”	与标题有关的词（照抄标题），换假标题就变	领域、年份（从标题推出）、引用数、是否认得出都读得出，且早于名字	不同问题下同一篇论文的表示在中间层相近，后期才按任务分开 **结果：**不高于 L4 文字水平，不支持	**M3 不支持；X3 显示中间层超出用词的语义表示，但不能与论文身份区分；H2 目前依据 M2 和 X1，其中 M2 的 Qwen 胜过 BERT 按补充 2 只作参考。** 另外，X4 最早可解码的顺序是第一作者（L8）早于年份（L40），与 H2 预测的方向相反。
H3 作者信息已在，只是不在词表方向	不一定	作者首 token 用 probe 读得出，lens 读不出	按作者训练的 probe 能跨任务迁移；作者专用 lens（M4）在中间层读得出名字	**文本基线不弱于 Qwen，不能排除文字解释，证据不足**（X4：名字首 token 上 BERT +0.088 ⩾ Qwen L8 +0.018；L8 信号主要在姓上，见上文 X4）

Hypotheses H1–H3 × readouts M1–M3 and current results (report, 2026-11-01)

Layer order (X5)

Earliest layer at which each kind of information appears · existing results only, each source keeps its own rule



Read early, carried, spoken late

≤L8–L16: title facts and the first-author token

L48–L52: the paper signal reaches the post-title tail (patching R ≥ 0.5 at L48, ≥ 0.95 at L52; knockout costs nothing from L52)

L53–L62: names readable by lens

Stage 12: where the paper information travels

All 408 GPU units done · true title → false_0 · share of the true–false gap recovered at the output

- **P1 one position at a time:** the colon carries at most 0.17 (L52)(Stage 12 P1, 1,186 papers, paper bootstrap; P1_patch_pos.csv); all name positions together 0.85 at L52(same; CI [0.830, 0.864]); the whole prompt 1.00 (= Stage 11 E1)(same, 0.996)
- **P2 one component of one layer:** attention at L48 +0.23 [0.22, 0.24](Stage 12 P2, 1,186 papers; P2_patch_comp.csv), MLP at L52–L56 +0.18 to +0.22(same) → attention reads the title in, MLP processes it next
- **N1 name plausibility:** without a title the true list looks "most real" in 0.255 of papers (chance 0.25) (Stage 12 N1, 8,894 R0/R1 papers; N1_prior_mc.csv); with the true title 0.625(same, CI [0.615, 0.635])
- **N2 pure name prior (no paper):** target − control +0.047 [0.031, 0.062] vs +0.247 with the true title(Stage 12 N2, final L63, 4,621 + 2,220, within bin; stage12_report.md) → prior ≈ 19% of the gap
- **F2 lens fitted on author prompts:** target − control ≤ 0 at L24–L56, +0.238 only at L62(Stage 12 F2, name positions, Stage 11 sample, true − 5 false; F2_author_lens.csv)

No single "notebook" token

The paper leaves the title at L44–L48 and is spread over the name positions; a lens fitted on author prompts still reads nothing paper-specific in the middle layers.

The Stage 11 timing holds for all 5 false titles

X6: replacement with each of the 5 false titles · X7: knockout minus the same knockout on the false-title run (S = T − F)

- **X6 replacement:** mean over 5 false titles 0.821 at L48 and 0.995 at L52(Stage 13 X6, 1,186 papers, team-cluster CI; x6_recovery.csv); each false title within 0.02 of false_0(x6_report.md) → "≈80% at L48, all from L52" is not specific to false_0
- **By name-token role:** surname first token 0.734 at L48, surname continuation 0.893; all roles ≥ 0.96 at L52(X6 roles, all papers; x6x7_roles.md)
- **X7 knockout:** blocking the title on the false-title run RAISES the true author score (F < 0), so the paper-specific loss S(L0) ≈ 1.00(Stage 13 X7, 1,186 papers, all: 1.004; x7_report.md); S(L52) = 0.029 [0.021, 0.038](same, all papers)
- **Pre-registered verdict:** holds for target (S(L52) 0.018) and all papers; control just misses (0.051, threshold 0.05)(X7, target 593 / control 593; x7_report.md) → control needs the title until L54–L56

What changes

Nothing in the timing: the title is used up to about L52 (control L54–56) and the paper information sits in the post-title positions by then.

Same title, different question

M3: probes across three questions (authors / year / field) · X1: patch A's title state from another question into B's author prompt

- **M3 transfer at the title:** $T \geq 0.92$ at L16–L40, but never above the L4 text level(Stage 13 M3, 9,108 papers, team folds; m3_report.md) (year $T - T(L4)$ −0.03 to −0.06)(same, CIs exclude 0) → M3 does not support H2
- **M3 at the colon:** $T < 0.5$ → the state is already organised by the question (H1)(same; summary_m3/m3_transfer.csv)
- **X1 replacement:** $R \approx 0.99$ (L24), 0.92 (L40), 0.42 (L48) whichever question the state came from(Stage 13 X1, test 582 pairs, team CI; x1_report.md); authors – year within ±0.02, except a small +0.019 at L48(same, [0.012, 0.026])
- **Retrieval (added 2026-11-05):** no layer retrieves the paper across questions better than L4(X5 retrieval, 9,108 papers; x5_retrieval.md; post-defence)

Reading

The title state hardly depends on the question, and the replacement effect is task-independent; but nothing goes beyond what the title text alone gives at L4.

Patchscope and paraphrases

X4: decode year / first author / field from the paper's own title state · X3: paraphrased title vs a word-matched other paper

- **X4 first author:** decodable from L8, D = +0.046 [0.035, 0.059](Stage 13 X4, test split, team CI; x4_report.md); year only from L40, +0.013 [0.008, 0.018](same); field never → order opposite to the H2 prediction
- **But:** the L8 signal sits on the surname (+0.066), where the text-baseline check failed(X4 role split, 1,178 papers; x4_rolesplit.csv) → may be explained by the title text
- **X3 paraphrase:** closer to the original than a word-matched paper beyond L4, peak L36 +0.16(Stage 13 X3, test 552 papers, G′ − G′(L4), team CI [0.146, 0.171]; x3_report.md); behaviour keeps ≈ 90% of the first-author preference(same, 0.90 [0.87, 0.94])
- **Post-hoc checks:** a sentence-embedding model without author knowledge shows a comparable advantage, and target = control(X3 text baselines and group split; x3_textbase_recovery.md, x3_groupsplit.md) → meaning, not separable from paper identity

H2 / H3 so far

H2 partly supported but not established (rests on M2 and X1); H3: insufficient evidence (text baseline not weaker than Qwen).

What we learned: read early, spoken late

Read early

Causally the title is read in L24–L50 and gathered at the name positions by L52, then no longer needed. A probe at the title token reads a little of it from L24 (re-checked).

Spoken late

At the name positions, paper-specific author knowledge becomes lens-readable only from ~L50, fastest over L57–L61.

No lens lead

Read with a lens at the names, target is not ahead of control in middle layers; the +0.21 gap is at the end (Stage 10, final L63, §6.4) (probe at the title: target − control +0.04–0.08, re-checked (Stage 13 E4b, first name token only)).

Real knowledge

Name prior is ~1/3 of the gap overall but ~0 on surnames (Stage 12 A, 0.087 / 0.247, §6.12); the rest is paper-specific and grows with citations.

Lenses agree

Jacobian lenses read more in middle layers, but nothing target-specific.

Name format

Initials carry little; full given names carry as much as surnames and help predict them.

Limits

Done: Stage 12

- Single-position patching (P1); attention vs MLP (P2, hook check passed)
- Name priors (N1, N2); author-prompt lens (F1, F2): middle layers still not readable

Done: Stage 13

- Probes read year, citations, field at the title from ≤L16 (M2); identity not above the L4 text level (M3)
- Replacement is task-independent (X1); paraphrase meaning beyond wording in L28-L40, not separable from paper identity (X3)
- First author decodable at L8, may be text-explainable (X4); 5 false titles confirm the Stage 11 timing (X6, X7)

Limits

- One model (Qwen3-32B)
- Teacher-forced preference, not generation
- Labels from the team lead's package; labelling code (elm) not available
- H2 (paper identity) partly supported but not established; H3 (hidden author knowledge): insufficient evidence

Added after the defence (2026-11-05)

Group comparisons (plan fixed before the results: addenda 7, 8) · CPU leftovers (post-hoc) · experiments still running

1. Group comparisons A / B / C / D: plan and status (results after the re-runs are approved)
2. CPU leftovers on existing data: X5 retrieval curve, M3 scores, N2 by layer, paper vs team intervals, M5
3. In progress (status only): X2 subspace patching, all-token probe, group re-runs

Added after the defence · groups

Group comparisons: plan and status

Plan fixed before any result (addenda 7, 8) · results go to the user and advisor first; not shown until approved

- **Groups:** open answer R0 / R1 / R2 × multiple-choice recognition strong / uncertain / fail; A = R0 × strong 4,110, B = R0 × fail 2,186, C = R1 × strong 511, D = R1 × fail 34 (cohort papers.parquet; addendum 7)
- **No circularity:** every comparison uses only the authors the model did NOT write; written authors are a reference column
- **Primary tests:** P1 recognition A − B; P2 partial recall C − A (first author / co-authors); P3 slope on orders right (0–4) in R0; within citation bin, team bootstrap, Holm over 4; confounder regressions beside each
- **Status:** CPU part computed and sent for review. Still running: intervention re-runs for R2 × strong (204 papers) and R0 × uncertain (600), and the E1 numerator / denominator by group (addendum 8; rerun_R2.parquet, rerun_U0_600.parquet)

Why no numbers here

User decision 2026-11-05: group results appear in the deck and report only after the re-runs and the E1 numerator / denominator are in and approved.

Added after the defence · CPU

CPU leftovers on existing data (post-hoc)

No new GPU runs · descriptive only · not in the pre-registration except the X5 retrieval curve

- **X5 retrieval curve (pre-registered, computed now):** top-1 retrieval of the same paper across questions is never above L4 (M3 cache, 9,108 papers; x5_retrieval.md) (title token 1.000 up to L44, 0.734 at the output) (same, team CI)
- **M3 behaviour scores:** year target − control +0.007 [0.003, 0.012]; first-author first token +0.147 [0.096, 0.191] (output, 4,621 / 2,220, within bin, team CI; m3_scores.md)
- **N2 by layer:** pure name prior target − control +0.047 at the output, +0.039 at logit L62; public J-lens L48 −0.133 (true title) (Stage 12 N2 rescored by layer; n2_layers.md) → only L62 reproduces the output pattern
- **Paper vs team intervals:** width ratio 0.97–1.00 (median); "excludes 0" changes in 2 of 1,155 rows (X1, X4, X6, X7, M3; v42_paper_vs_team.md)
- **M5 true vs false title at the name positions:** output 0.919; logit L24 0.516 → L62 0.899 (share of pairs, all papers × 5 false titles, team CI; m5_report.md); probe at title / colon AUROC 0.50–0.52 (same, Part B)

Added after the defence · running

Still running (status only, no results)

All pre-registered before any data: addenda 6, 8 and 9 (sha256 in the report)

- **X2 subspace patching (addendum 6):** subspaces fitted; GPU shard 0 of 24 done and checked; the rest is queued behind the group re-runs
- **All-token probe (addendum 9):** the title-position probe (E4b) so far scores only the FIRST name token, mostly an initial; the new version scores every name token, split into initial / given name / surname. GPU data step (~1 A100-hour) queued after the re-runs and X2
- **Group re-runs (addendum 8):** E1 / E2 (and P1 / P2 for R2) for R2 × strong and R0 × uncertain; E1 numerator / denominator by group